var, x, y, z
$label$, L, $Next$, $Done$
lit
f
$dsort$, D
$tvar$, a, b
$index$, i, j, k, n

terminals	::=		
	|	$\rightarrow$	
	|	$\rightsquigarrow$	
	|	$\rightarrow$	
	|	$\mid$	
	|	$\vee$	
	|	@	
	|	$\multimap$	
	|	$\wedge$	
	|	$\Rightarrow$	
	|	$\in$	
	|	true	
	|	false	
blit	::=		boolean literals
	|	true	
	|	false	
$A,\ B,\ C$	::=		types
	|	int	integers
	|	bool	booleans
	|	ptr A	pointer
	|	$(A_1, .., A_n)$	record
	|	$D(A_1, .., A_n)$	datatype application
	|	a	type variable
$\overline{A}$	::=		type lists
	|	$[]$	
	|	$A{::}\overline{A}$	
$\mathit{eff},\ \epsilon$	::=		effects
	|	**0**	
	|	**1**	
	|	**S**	
τ	::=		sorts
	|	int	integers
	|	bool	booleans
	|	ptr τ'	pointer
	|	$(\tau_1, .., \tau_n)$	tuple
	|	$D(\tau_1, .., \tau_n)$	datasort application
	|	pred τ'	predicate sort
	|	a	type variable
F	::=		function signatures
	|	$\tau \rightarrow_{\mathit{eff}} \tau'$	
Γ	::=		contexts
	|	$\cdot$	

	$\mid$	$\Gamma, x{:}_{\mathit{eff}}\tau$	
	$\mid$	Γ, Γ'	
Σ_c	$::=$		core signatures
	$\mid$	$\cdot$	
	$\mid$	$\Sigma_c, f : F$	
	$\mid$	$\Sigma_c, \mathsf{fun}\ f(x : \tau) \rightarrow_{\mathit{eff}} \tau' = ce$	
	$\mid$	$\Sigma_c, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}$	
	$\mid$	$\Sigma_c, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}$	
Σ_s	$::=$		surface signatures
	$\mid$	$\cdot$	
	$\mid$	$\Sigma_s, f : F$	
	$\mid$	$\Sigma_s, \mathsf{fun}\ f : \tau \rightarrow_{\mathit{eff}} \tau' = \{\overline{\mathbf{pat} \rightarrow \mathbf{se}}\}$	
	$\mid$	$\Sigma_s, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}$	
	$\mid$	$\Sigma_s, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}$	
Σ_r	$::=$		refined signatures
	$\mid$	$\cdot$	
	$\mid$	$\Sigma_r, f : F$	
	$\mid$	$\Sigma_r, \mathsf{fun}\ f(x : \tau) \rightarrow_{\mathit{eff}} \tau' = ce$	
	$\mid$	$\Sigma_r, f : RF$	
	$\mid$	$\Sigma_r, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}$	
	$\mid$	$\Sigma_r, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}$	
Φ	$::=$		type variable contexts
	$\mid$	$\cdot$	
	$\mid$	Φ, a	
	$\mid$	$\overline{a}$	
prim	$::=$		primitives
	$\mid$	$\mathbf{Add}$	addition
	$\mid$	$\mathbf{Sub}$	subtraction
	$\mid$	$\mathbf{Mul}$	multiplication
	$\mid$	$\mathbf{Div}$	division
	$\mid$	$\mathbf{And}$	conjunction
	$\mid$	$\mathbf{Or}$	disjunction
	$\mid$	$\mathbf{Not}$	negation
	$\mid$	$\mathbf{Lt}$	less than
	$\mid$	$\mathbf{Le}$	less or equal
	$\mid$	$\mathbf{Gt}$	greater than
	$\mid$	$\mathbf{Ge}$	greater or equal
	$\mid$	$\mathbf{Eq}\,[A]$	equality test
	$\mid$	$\mathbf{New}\,[A]$	allocate
	$\mid$	$\mathbf{Del}\,[A]$	deallocate
	$\mid$	$\mathbf{Get}\,[A]$	dereference
	$\mid$	$\mathbf{Set}\,[A]$	update
	$\mid$	$\mathbf{Own}\,[A]$	ownership

ce	$::=$		core expressions
	$\mid$	x	
	$\mid$	lit	integer literal
	$\mid$	$blit$	boolean literal
	$\mid$	$\mathbf{let}\, x = ce_1; ce_2$	
	$\mid$	$(ce_1, .., ce_n)$	
	$\mid$	$\mathbf{let}\, (x_1, .., x_n) = ce_1; ce_2$	
	$\mid$	$L\, ce$	
	$\mid$	$\mathbf{case}\, ce\, \mathbf{of}\, \{L_1\, x_1 \to ce_1 \mid .. \mid L_n\, x_n \to ce_n\}$	
	$\mid$	$\mathbf{iter}\, (x = ce_1)\{ce_2\}$	
	$\mid$	$prim\, ce$	
	$\mid$	$\mathbf{if}\, ce_1\, \mathbf{then}\, ce_2\, \mathbf{else}\, ce_3$	conditional
	$\mid$	$ce : \tau$	
	$\mid$	$f\, ce$	
	$\mid$	$ce_1 \doteq ce_2$	
	$\mid$	$\mathbf{take}\, x = ce_1; ce_2$	
	$\mid$	$\mathbf{return}\, ce$	
	$\mid$	(ce)	
	$\mid$	$[ce/x]ce'$	
se	$::=$		surface expressions
	$\mid$	x	
	$\mid$	lit	integer literal
	$\mid$	$blit$	boolean literal
	$\mid$	$\mathbf{let}\, pat = se_1; se_2$	
	$\mid$	$(se_1, .., se_n)$	
	$\mid$	$L\, se$	
	$\mid$	$\mathbf{case}\, se\, \mathbf{of}\, \{pat_1 \to se_1 \mid .. \mid pat_n \to se_n\}$	
	$\mid$	$\mathbf{iter}\, (pat = se_1)\{se_2\}$	
	$\mid$	$prim\, se$	
	$\mid$	$\mathbf{if}\, se_1\, \mathbf{then}\, se_2\, \mathbf{else}\, se_3$	conditional
	$\mid$	$se : \tau$	
	$\mid$	$f\, se$	
	$\mid$	$se_1 \doteq se_2$	
	$\mid$	$\mathbf{take}\, pat = se_1; se_2$	
	$\mid$	$\mathbf{return}\, se$	
$prog$	$::=$		surface programs
	$\mid$	$\mathsf{fun}\, f : \tau \to_{\mathit{eff}} \tau' = \{\overline{\mathbf{pat} \to \mathbf{se}}\} \quad prog$	
	$\mid$	$\mathbf{sort}\, D(a_1, .., a_i) = \{L_1 : \tau_1 \mid .. \mid L_n : \tau_n\} prog$	
	$\mid$	$\mathbf{type}\, D(a_1, .., a_i) = \{L_1 : A_1 \mid .. \mid L_n : A_n\} prog$	
	$\mid$	$\mathsf{main} : \tau_{\mathit{eff}} = se$	
$cprog$	$::=$		core programs
	$\mid$	$\mathsf{fun}\, f(x : \tau) \to_{\mathit{eff}} \tau' = ce \quad cprog$	
	$\mid$	$\mathbf{sort}\, D(a_1, .., a_i) = \{L_1 : \tau_1 \mid .. \mid L_n : \tau_n\} cprog$	
	$\mid$	$\mathbf{type}\, D(a_1, .., a_i) = \{L_1 : A_1 \mid .. \mid L_n : A_n\} cprog$	
	$\mid$	$\mathsf{main} : \tau_{\mathit{eff}} = ce$	
$rprog$	$::=$		refined programs

	$\mid$	$\textbf{sort}\, D(a_1, .., a_i) = \{L_1 : \tau_1 \mid .. \mid L_n : \tau_n\} rprog$	
	$\mid$	$\textbf{type}\, D(a_1, .., a_i) = \{L_1 : A_1 \mid .. \mid L_n : A_n\} rprog$	
	$\mid$	$\mathsf{fun}\, f(x : \tau) \rightarrow_{\mathit{eff}} \tau' = ce \quad rprog$	
	$\mid$	$\mathsf{fun}\, f(Pf_1) \rightarrow_{\mathit{eff}} Pf_2 = crt \quad rprog$	
	$\mid$	$\mathsf{main} : Pf_{\mathit{eff}} = crt$	
pat	$::=$		patterns
	$\mid$	x	
	$\mid$	$L\, pat$	
	$\mid$	$(pat_1, .., pat_n)$	
β	$::=$		pattern bindings
	$\mid$	$\cdot$	
	$\mid$	$pat : \tau, \beta$	
	$\mid$	$pat_1 : \tau_1, .., pat_n : \tau_n, \beta$	
	$\mid$	$pat : \tau$	
$\mathcal{E}$	$::=$		evaluation contexts
	$\mid$	$[\cdot]$	
	$\mid$	$\textbf{let}\, x = ce; \mathcal{E}$	
$\overline{y}$	$::=$		scrutinee variables
	$\mid$	$\cdot$	
	$\mid$	$y, \overline{y}$	
	$\mid$	$y_1, .., y_n, \overline{y}$	
M	$::=$		match branches
	$\mid$	$\cdot$	
	$\mid$	$\beta \rightarrow [\Gamma][\mathcal{E}]se \mid M$	
	$\mid$	$\beta \rightarrow [\Gamma][\mathcal{E}]se$	
$formula$	$::=$		
	$\mid$	$judgement$	
	$\mid$	$x{:}_{\mathit{eff}}\tau \in \Gamma$	
	$\mid$	$f{:}\tau \rightarrow_{\mathit{eff}} \tau' \in \Sigma_c$	
	$\mid$	$A\ \mathsf{eqtype}$	
	$\mid$	$\textbf{sort}\, D(a_1, .., a_i)\, \textbf{in}\, \Sigma_c$	
	$\mid$	$L{:}\tau \in \mathrm{ctors}(D) \in \Sigma_c$	
	$\mid$	$\tau = \textbf{subst}\, \tau_1/a_1, .., \tau_i/a_i\, \textbf{in}\, \tau'$	
	$\mid$	$\textbf{sort}\, D(a_1, .., a_k) = \{L_1 : \tau_1 \mid .. \mid L_n : \tau_n\}\, \textbf{in}\, \Sigma_c$	
	$\mid$	$\textbf{type}\, D(a_1, .., a_i)\, \textbf{in}\, \Sigma_c$	
	$\mid$	$L{:}A \in \mathsf{tyctors}(D) \in \Sigma_c$	
	$\mid$	$A = \textbf{subst}\, A_1/a_1, .., A_i/a_i\, \textbf{in}\, A'$	
	$\mid$	$\textbf{type}\, D(a_1, .., a_k) = \{L_1 : A_1 \mid .. \mid L_n : A_n\}\, \textbf{in}\, \Sigma_c$	
	$\mid$	$\mathsf{type}\, D(a, b) = \{\mathsf{Next}{:}a \mid \mathsf{Done}{:}b\} \in \Sigma_c$	
	$\mid$	$a \in \Phi$	
	$\mid$	$L \neq L'$	
	$\mid$	$M = \overline{pat{:}\tau, \cdot \rightarrow [\cdot][\cdot]\, se}$	
	$\mid$	$ce = \mathcal{E}[ce']$	

	$\mid$	$D \neq D'$
	$\mid$	$x \neq y$
	$\mid$	$n = k$
	$\mid$	$\mathit{eff} = \mathit{eff}_1 \vee \mathit{eff}_2$
	$\mid$	$\mathit{eff} = \mathit{eff}_1 \vee \mathit{eff}_2 \vee \mathit{eff}_3$
	$\mid$	$\mathit{eff} = \mathit{eff}_1 \vee .. \vee \mathit{eff}_n$
	$\mid$	$\Gamma = x_1 : \tau_1[\mathit{eff}_1], .., x_n : \tau_n[\mathit{eff}_n]$
	$\mid$	$\mathit{formula}_1 \quad .. \quad \mathit{formula}_n$
	$\mid$	$x{:}ce\ [\mathsf{log}] \in \Delta$
	$\mid$	$x{:}ce@ce'\ [\mathsf{res}(u)] \in \Delta$
	$\mid$	$x{:}_{\mathit{eff}}\tau \in \Delta$
	$\mid$	$u \in \{1, ?\}$
	$\mid$	$u \in \{0, ?\}$
	$\mid$	$u = u_1 \sqcap u_2$
	$\mid$	$\Delta' = \mathsf{affinize}(\Delta)$
	$\mid$	$\mathsf{zero}(\Delta)$
	$\mid$	$\Gamma = \lvert\Delta\rvert$
	$\mid$	$\Gamma = \mathsf{Bind}(\cdot; Pf)$
	$\mid$	$\mathcal{C} = \Delta \Rightarrow \mathcal{C}'$
	$\mid$	$f(\mathbf{x} : \tau) \rightarrow_{\mathit{eff}} \tau' = ce \in \Sigma_c$
	$\mid$	$\mathit{eff}' \in \{\mathbf{S}, \lfloor \mathit{eff} \rfloor\}$
	$\mid$	$f{:}RF \in \Sigma_c$
	$\mid$	$f{:}RF \in \Sigma_r$
	$\mid$	$f{:}\tau \rightarrow_{\mathit{eff}} \tau' \in \Sigma_r$
	$\mid$	$f(\mathbf{x} : \tau) \rightarrow_{\mathit{eff}} \tau' = ce \in \Sigma_r$
	$\mid$	$\mathsf{len}(\Delta) = \mathsf{len}(\Delta')$
	$\mid$	$A = \mathsf{Prod}(\overline{A})$
u	$::=$	usage flags
	$\mid$	0
	$\mid$	1
	$\mid$	$?$
Δ	$::=$	refined contexts
	$\mid$	$\cdot$
	$\mid$	$\Delta, x{:}_{\mathit{eff}}\tau$
	$\mid$	$\Delta, x{:}ce\ [\mathsf{log}]$
	$\mid$	$\Delta, x{:}ce@ce'\ [\mathsf{res}(u)]$
	$\mid$	Δ, Δ'
$\overline{\Delta}$	$::=$	context lists
	$\mid$	$\Delta; \overline{\Delta}$
	$\mid$	$\cdot$
	$\mid$	$(\overline{\Delta})$
	$\mid$	$\Delta_1; .. ; \Delta_n$
$\mathcal{C}$	$::=$	SMT constraints
	$\mid$	$\top$
	$\mid$	$\mathcal{C}_1 \wedge \mathcal{C}_2$
	$\mid$	$\forall x{:}\tau.\, \mathcal{C}$

	$\mid$	$ce \Rightarrow \mathcal{C}$	
	$\mid$	ce	
	$\mid$	$\bot$	
	$\mid$	$(\mathcal{C})$	
Pf	$::=$		proof sorts
	$\mid$	$\cdot$	
	$\mid$	$x{:}_{\mathit{eff}}\tau, Pf$	
	$\mid$	$x{:}ce\ [\mathsf{log}], Pf$	
	$\mid$	$x{:}ce@ce'\ [\mathsf{res}], Pf$	
	$\mid$	$[ce/x]Pf$	
RF	$::=$		refined function types
	$\mid$	$Pf_1 \multimap Pf_2\ [\mathit{eff}]$	
q	$::=$		refined patterns
	$\mid$	$()$	
	$\mid$	(x, q)	
crt	$::=$		core refined terms
	$\mid$	x	
	$\mid$	lit	integer literal
	$\mid$	$blit$	boolean literal
	$\mid$	$\mathbf{let}\ q = crt_1; crt_2$	
	$\mid$	$crt : Pf$	
	$\mid$	$prim\ rsp$	
	$\mid$	$f\ rsp$	
	$\mid$	(rsp)	
	$\mid$	$\mathbf{iter}\,[ce](q = crt_1)\{crt_2\}$	
	$\mid$	$\mathbf{if}\,[x]ce\,\mathbf{then}\,crt_1\,\mathbf{else}\,crt_2$	
	$\mid$	$\mathbf{case}\,[y]ce\,\mathbf{of}\,\{L_1\,x_1 \to crt_1 \mid .. \mid L_n\,x_n \to crt_n\}$	
	$\mid$	$\mathsf{exfalso}$	
	$\mid$	$\mathsf{open\text{-}take}\ rpf$	
rsp	$::=$		refined spines
	$\mid$	$\cdot$	
	$\mid$	ce, rsp	
	$\mid$	rpf, rsp	
	$\mid$	lpf, rsp	
lpf	$::=$		logical proof facts
	$\mid$	x	
	$\mid$	auto	
	$\mid$	$\mathsf{unfold}\ f(ce)$	
	$\mid$	$\mathsf{open\text{-}ret}\ rpf$	
	$\mid$	$lpf : ce$	
rpf	$::=$		resource proof facts
	$\mid$	x	

	$\mid$	make-ret lpf
	$\mid$	make-take crt
	$\mid$	$rpf : ce@ce'$
$Jtype$	$::=$	
	$\mid$	$\Sigma_c; \Phi \vdash A$ wf
	$\mid$	$\Sigma_c; \Phi; D' \vdash A$ guarded
	$\mid$	$\Sigma_c \vdash$ sort $D(\overline{a}) = \{\overline{L : \tau}\}$ ok
	$\mid$	$\Sigma_c \vdash$ type $D(\overline{a}) = \{\overline{L : A}\}$ ok
	$\mid$	$\Sigma_c; \Phi \vdash \tau$ wf
	$\mid$	A eqtype
	$\mid$	$\mathit{eff} \leq \mathit{eff}'$
	$\mid$	$\mathit{eff}' = \lfloor \mathit{eff} \rfloor$
	$\mid$	$\mathit{eff} = \mathit{eff}_1 \vee \mathit{eff}_2$
	$\mid$	$prim{:}\tau \rightarrow_{\mathit{eff}} \tau'$
	$\mid$	$\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce \Rightarrow \tau$
	$\mid$	$\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce \Leftarrow \tau$
	$\mid$	$\Sigma_c \vdash cprog$
$Jcore$	$::=$	
	$\mid$	$L : \tau \,\mathbf{in}\, D(\tau_1, .., \tau_i) \,\mathbf{in}\, \Sigma_c$
	$\mid$	$D(a_1, .., a_k) = \{L_1 : \tau_1 \mid .. \mid L_n : \tau_n\} \in \Sigma_c$
$Jelaborate$	$::=$	
	$\mid$	$\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se \Rightarrow \tau \rightsquigarrow ce$
	$\mid$	$\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se \Leftarrow \tau \rightsquigarrow ce$
	$\mid$	$\Sigma_s \vdash prog$ $\rightsquigarrow \Sigma_c \vdash cprog$
$Jmatch$	$::=$	
	$\mid$	M has_con
	$\mid$	M has_tup
	$\mid$	$M/var \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'$
	$\mid$	$M/L{:}\tau \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'$
	$\mid$	$M/\mathbf{tup} \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'$
	$\mid$	$\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \overline{y}; M \Leftarrow_\epsilon \tau \rightsquigarrow ce$
	$\mid$	$\mathcal{E}[ce] \rightsquigarrow ce'$
$Jrefine$	$::=$	
	$\mid$	$\Sigma_r \vdash \Delta$ wf
	$\mid$	$\Gamma = \lvert\Delta\rvert$
	$\mid$	$\Delta' = \mathsf{affinize}(\Delta)$
	$\mid$	$\mathsf{zero}(\Delta)$
	$\mid$	$\mathcal{C} = \Delta \Rightarrow \mathcal{C}'$
	$\mid$	$\Sigma_r; \Gamma \vdash \mathcal{C}$ wf
	$\mid$	$\Sigma_r; \Gamma \vdash_{\mathit{eff}} Pf$ wf
	$\mid$	$\Sigma_r \vdash RF$ wf
	$\mid$	$\Sigma_r \vdash f{:}RF$
	$\mid$	$\overline{A} = \{Pf\}$
	$\mid$	$\Delta = \mathsf{PfToCtx}(\Delta'; Pf)$
	$\mid$	$\Sigma_c = \mathsf{Comp}(\Sigma_r)$

$\mid$	$\vdash \Sigma_r$ wf
$\mid$	$\Sigma_r; \Delta \vdash lpf \Rightarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C}$
$\mid$	$\Sigma_r; \Delta \vdash lpf \Leftarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C}$
$\mid$	$\Sigma_r; \Delta \vdash x : ce@ce' \dashv \Delta'$
$\mid$	$\Sigma_r; \Delta \vdash rpf \Rightarrow ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}$
$\mid$	$\Sigma_r; \Delta \vdash rpf \Leftarrow ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}$
$\mid$	$\Sigma_r; \Delta \vdash Pf_1 = Pf_2 \rightsquigarrow \mathcal{C}$
$\mid$	$\Sigma_r \vdash \Delta_0 \sqcap \Delta_1 = \Delta$
$\mid$	$\Sigma_r; \Gamma \vdash q : Pf \dashv \Delta$
$\mid$	$\Sigma_r; \Delta \vdash_{\mathit{eff}} crt \Rightarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}$
$\mid$	$\Sigma_r; \Delta \vdash_{\mathit{eff}} y{:}ce$ is $L\,x \rightarrow crt \Leftarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}$
$\mid$	$\Sigma_r; \Delta \vdash_{\mathit{eff}} crt \Leftarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}$
$\mid$	$\Sigma_r; \Delta \vdash_{\mathit{eff}} rsp : RF \gg Pf \dashv \Delta' \rightsquigarrow \mathcal{C}$
$\mid$	$\Sigma_r; \Delta \vdash_{\mathit{eff}} rsp : Pf \dashv \Delta' \rightsquigarrow \mathcal{C}$
$\mid$	$\Sigma_r \vdash rprog \dashv \Sigma_r' \rightsquigarrow \mathcal{C}$
$\mid$	$\Sigma_r \vdash \sqcap\overline{\Delta} = \Delta$

judgement	::=	
	$\mid$	*Jtype*
	$\mid$	*Jcore*
	$\mid$	*Jelaborate*
	$\mid$	*Jmatch*
	$\mid$	*Jrefine*

user_syntax	::=	
	$\mid$	*var*
	$\mid$	*label*
	$\mid$	*lit*
	$\mid$	*f*
	$\mid$	*dsort*
	$\mid$	*tvar*
	$\mid$	*index*
	$\mid$	*terminals*
	$\mid$	*blit*
	$\mid$	A
	$\mid$	$\overline{A}$
	$\mid$	*eff*
	$\mid$	τ
	$\mid$	F
	$\mid$	Γ
	$\mid$	Σ_c
	$\mid$	Σ_s
	$\mid$	Σ_r
	$\mid$	Φ
	$\mid$	*prim*
	$\mid$	*ce*
	$\mid$	*se*
	$\mid$	*prog*
	$\mid$	*cprog*
	$\mid$	*rprog*

$$
\begin{array}{|l}
pat \\
\beta \\
\mathcal{E} \\
\overline{y} \\
M \\
formula \\
u \\
\Delta \\
\overline{\Delta} \\
\mathcal{C} \\
Pf \\
RF \\
q \\
crt \\
rsp \\
lpf \\
rpf
\end{array}
$$

$\boxed{\Sigma_c; \Phi \vdash A \ \mathsf{wf}}$

$$\frac{}{\Sigma_c; \Phi \vdash \mathsf{int\ wf}} \quad \textsc{TWf_int}$$

$$\frac{}{\Sigma_c; \Phi \vdash \mathsf{bool\ wf}} \quad \textsc{TWf_bool}$$

$$\frac{\Sigma_c; \Phi \vdash A \ \mathsf{wf}}{\Sigma_c; \Phi \vdash \mathsf{ptr}\ A \ \mathsf{wf}} \quad \textsc{TWf_ptr}$$

$$\frac{\Sigma_c; \Phi \vdash A_1 \ \mathsf{wf} \quad .. \quad \Sigma_c; \Phi \vdash A_n \ \mathsf{wf}}{\Sigma_c; \Phi \vdash (A_1, \ .., A_n) \ \mathsf{wf}} \quad \textsc{TWf_record}$$

$$\frac{\begin{array}{c}\mathbf{type}\, D(a_1, \ .., a_k)\, \mathbf{in}\, \Sigma_c \quad n = k \\ \Sigma_c; \Phi \vdash A_1 \ \mathsf{wf} \quad .. \quad \Sigma_c; \Phi \vdash A_n \ \mathsf{wf}\end{array}}{\Sigma_c; \Phi \vdash D(A_1, \ .., A_n) \ \mathsf{wf}} \quad \textsc{TWf_app}$$

$$\frac{a \in \Phi}{\Sigma_c; \Phi \vdash a \ \mathsf{wf}} \quad \textsc{TWf_tvar}$$

$\boxed{\Sigma_c; \Phi; D' \vdash A \ \mathsf{guarded}}$

$$\frac{}{\Sigma_c; \Phi; D' \vdash \mathsf{int\ guarded}} \quad \textsc{TWfG_int}$$

$$\frac{}{\Sigma_c; \Phi; D' \vdash \mathsf{bool\ guarded}} \quad \textsc{TWfG_bool}$$

$$\frac{\Sigma_c; \Phi \vdash A \ \mathsf{wf}}{\Sigma_c; \Phi; D' \vdash \mathsf{ptr}\ A \ \mathsf{guarded}} \quad \textsc{TWfG_ptr}$$

$$\frac{\Sigma_c; \Phi; D' \vdash A_1 \ \mathsf{guarded} \quad .. \quad \Sigma_c; \Phi; D' \vdash A_n \ \mathsf{guarded}}{\Sigma_c; \Phi; D' \vdash (A_1, \ .., A_n) \ \mathsf{guarded}} \quad \textsc{TWfG_record}$$

$$\frac{\begin{array}{c}\mathbf{type}\, D(a_1, \ .., a_k)\, \mathbf{in}\, \Sigma_c \quad n = k \quad D \neq D' \\ \Sigma_c; \Phi \vdash A_1 \ \mathsf{wf} \quad .. \quad \Sigma_c; \Phi \vdash A_n \ \mathsf{wf}\end{array}}{\Sigma_c; \Phi; D' \vdash D(A_1, \ .., A_n) \ \mathsf{guarded}} \quad \textsc{TWfG_app}$$

$$\frac{a \in \Phi}{\Sigma_c; \Phi; D' \vdash a \ \mathsf{guarded}} \quad \textsc{TWfG_tvar}$$

$\boxed{\Sigma_c \vdash \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}\ \mathsf{ok}}$

$$\frac{\Sigma_c, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}; \overline{a} \vdash \tau_1\ \mathsf{wf} \quad .. \quad \Sigma_c, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}; \overline{a} \vdash \tau_n\ \mathsf{wf}}{\Sigma_c \vdash \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}\ \mathsf{ok}} \quad \text{SDWF_OK}$$

$\boxed{\Sigma_c \vdash \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}\ \mathsf{ok}}$

$$\frac{\Sigma_c, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}; \overline{a}; D \vdash A_1\ \mathsf{guarded} \quad .. \quad \Sigma_c, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}; \overline{a}; D \vdash A_n\ \mathsf{guarded}}{\Sigma_c \vdash \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}\ \mathsf{ok}} \quad \text{TDWF_OK}$$

$\boxed{\Sigma_c; \Phi \vdash \tau\ \mathsf{wf}}$

$$\frac{}{\Sigma_c; \Phi \vdash \mathsf{int}\ \mathsf{wf}} \quad \text{SWF_INT}$$

$$\frac{}{\Sigma_c; \Phi \vdash \mathsf{bool}\ \mathsf{wf}} \quad \text{SWF_BOOL}$$

$$\frac{\Sigma_c; \Phi \vdash \tau\ \mathsf{wf}}{\Sigma_c; \Phi \vdash \mathsf{ptr}\ \tau\ \mathsf{wf}} \quad \text{SWF_PTR}$$

$$\frac{\Sigma_c; \Phi \vdash \tau_1\ \mathsf{wf} \quad .. \quad \Sigma_c; \Phi \vdash \tau_n\ \mathsf{wf}}{\Sigma_c; \Phi \vdash (\tau_1, .., \tau_n)\ \mathsf{wf}} \quad \text{SWF_RECORD}$$

$$\frac{\begin{array}{c}\mathbf{sort}\ D(a_1, .., a_k)\ \mathbf{in}\ \Sigma_c \quad n = k \\ \Sigma_c; \Phi \vdash \tau_1\ \mathsf{wf} \quad .. \quad \Sigma_c; \Phi \vdash \tau_n\ \mathsf{wf}\end{array}}{\Sigma_c; \Phi \vdash D(\tau_1, .., \tau_n)\ \mathsf{wf}} \quad \text{SWF_APP}$$

$$\frac{\Sigma_c; \Phi \vdash \tau\ \mathsf{wf}}{\Sigma_c; \Phi \vdash \mathsf{pred}\ \tau\ \mathsf{wf}} \quad \text{SWF_PRED}$$

$$\frac{a \in \Phi}{\Sigma_c; \Phi \vdash a\ \mathsf{wf}} \quad \text{SWF_TVAR}$$

$\boxed{A\ \mathsf{eqtype}}$

$$\frac{}{\mathsf{int\ eqtype}} \quad \text{EQTY_INT}$$

$$\frac{}{\mathsf{bool\ eqtype}} \quad \text{EQTY_BOOL}$$

$$\frac{}{\mathsf{ptr}\ A\ \mathsf{eqtype}} \quad \text{EQTY_PTR}$$

$\boxed{\mathit{eff} \leq \mathit{eff}'}$

$$\frac{}{\mathbf{0} \leq \mathbf{0}} \quad \text{SUB_PURE_PURE}$$

$$\frac{}{\mathbf{0} \leq \mathbf{1}} \quad \text{SUB_PURE_IMPURE}$$

$$\frac{}{\mathbf{1} \leq \mathbf{1}} \quad \text{SUB_IMPURE_IMPURE}$$

$$\frac{}{\mathbf{0} \leq \mathbf{S}} \quad \text{SUB_PURE_SPEC}$$

$$\frac{}{\mathbf{S} \leq \mathbf{S}} \quad \text{SUB_SPEC_SPEC}$$

$\boxed{\mathit{eff}' = \lfloor \mathit{eff} \rfloor}$

$$\frac{}{\mathbf{0} = \lfloor \mathbf{0} \rfloor}\ \textsc{Pur_pure}$$

$$\frac{}{\mathbf{0} = \lfloor \mathbf{1} \rfloor}\ \textsc{Pur_impure}$$

$$\frac{}{\mathbf{S} = \lfloor \mathbf{S} \rfloor}\ \textsc{Pur_spec}$$

$\boxed{\mathit{eff} = \mathit{eff}_1 \vee \mathit{eff}_2}$

$$\frac{}{\mathit{eff} = \mathit{eff} \vee \mathit{eff}}\ \textsc{Join_refl}$$

$$\frac{}{\mathbf{1} = \mathbf{0} \vee \mathbf{1}}\ \textsc{Join_pure_impure}$$

$$\frac{}{\mathbf{1} = \mathbf{1} \vee \mathbf{0}}\ \textsc{Join_impure_pure}$$

$$\frac{}{\mathbf{S} = \mathbf{0} \vee \mathbf{S}}\ \textsc{Join_pure_spec}$$

$$\frac{}{\mathbf{S} = \mathbf{S} \vee \mathbf{0}}\ \textsc{Join_spec_pure}$$

$\boxed{\mathit{prim}{:}\tau \rightarrow_{\mathit{eff}} \tau'}$

$$\frac{}{\mathbf{Add}{:}(\mathsf{int}, \mathsf{int}) \rightarrow_{\mathbf{0}} \mathsf{int}}\ \textsc{PT_add}$$

$$\frac{}{\mathbf{Sub}{:}(\mathsf{int}, \mathsf{int}) \rightarrow_{\mathbf{0}} \mathsf{int}}\ \textsc{PT_sub}$$

$$\frac{}{\mathbf{Mul}{:}(\mathsf{int}, \mathsf{int}) \rightarrow_{\mathbf{0}} \mathsf{int}}\ \textsc{PT_mul}$$

$$\frac{}{\mathbf{Div}{:}(\mathsf{int}, \mathsf{int}) \rightarrow_{\mathbf{1}} \mathsf{int}}\ \textsc{PT_div}$$

$$\frac{}{\mathbf{And}{:}(\mathsf{bool}, \mathsf{bool}) \rightarrow_{\mathbf{0}} \mathsf{bool}}\ \textsc{PT_and}$$

$$\frac{}{\mathbf{Or}{:}(\mathsf{bool}, \mathsf{bool}) \rightarrow_{\mathbf{0}} \mathsf{bool}}\ \textsc{PT_or}$$

$$\frac{}{\mathbf{Not}{:}\mathsf{bool} \rightarrow_{\mathbf{0}} \mathsf{bool}}\ \textsc{PT_not}$$

$$\frac{}{\mathbf{Lt}{:}(\mathsf{int}, \mathsf{int}) \rightarrow_{\mathbf{0}} \mathsf{bool}}\ \textsc{PT_lt}$$

$$\frac{}{\mathbf{Le}{:}(\mathsf{int}, \mathsf{int}) \rightarrow_{\mathbf{0}} \mathsf{bool}}\ \textsc{PT_le}$$

$$\frac{}{\mathbf{Gt}{:}(\mathsf{int}, \mathsf{int}) \rightarrow_{\mathbf{0}} \mathsf{bool}}\ \textsc{PT_gt}$$

$$\frac{}{\mathbf{Ge}{:}(\mathsf{int}, \mathsf{int}) \rightarrow_{\mathbf{0}} \mathsf{bool}}\ \textsc{PT_ge}$$

$$\frac{A\ \mathsf{eqtype}}{\mathbf{Eq}\,[A]{:}(A, A) \rightarrow_{\mathbf{0}} \mathsf{bool}}\ \textsc{PT_eq}$$

$$\frac{}{\mathbf{New}\,[A]{:}A \rightarrow_{\mathbf{1}} \mathsf{ptr}\ A}\ \textsc{PT_new}$$

$$\frac{}{\mathbf{Del}\,[A]{:}\mathsf{ptr}\ A \rightarrow_{\mathbf{1}} (\,)}\ \textsc{PT_del}$$

$$\frac{}{\mathbf{Get}\,[A]{:}\mathsf{ptr}\ A \rightarrow_{\mathbf{1}} A}\ \textsc{PT_get}$$

$$\frac{}{\textbf{Set}\,[A]{:}(\mathsf{ptr}\ A, A) \rightarrow_{\mathbf{1}} ()} \quad \text{PT_SET}$$

$$\frac{}{\textbf{Own}\,[A]{:}\mathsf{ptr}\ A \rightarrow_{\mathbf{S}} \mathsf{pred}\ (A)} \quad \text{PT_OWN}$$

$\boxed{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{ce} \Rightarrow \tau}$

$$\frac{}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{lit} \Rightarrow \mathsf{int}} \quad \text{SYN_INT}$$

$$\frac{}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{blit} \Rightarrow \mathsf{bool}} \quad \text{SYN_BOOL}$$

$$\frac{x{:}_{\mathit{eff}'}\tau \in \Gamma \quad \mathit{eff}' \leq \mathit{eff}_0}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} x \Rightarrow \tau} \quad \text{SYN_VAR}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{ce} \Leftarrow \tau}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{ce} : \tau \Rightarrow \tau} \quad \text{SYN_ANNOT}$$

$$\frac{\begin{array}{l} \mathit{prim}{:}\tau \rightarrow_{\mathit{eff}'} \tau' \\ \mathit{eff}' \leq \mathit{eff}_0 \\ \mathit{eff}'' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}''} \mathit{ce} \Leftarrow \tau \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{prim}\ \mathit{ce} \Rightarrow \tau'} \quad \text{SYN_APP}$$

$$\frac{\begin{array}{l} f{:}\tau \rightarrow_{\mathit{eff}'} \tau' \in \Sigma_c \\ \mathit{eff}' \leq \mathit{eff}_0 \\ \mathit{eff}'' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}''} \mathit{ce} \Leftarrow \tau \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} f\ \mathit{ce} \Rightarrow \tau'} \quad \text{SYN_CALL}$$

$$\frac{\begin{array}{l} A\ \mathsf{eqtype} \\ \mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} \mathit{ce}_1 \Rightarrow A \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} \mathit{ce}_2 \Leftarrow A \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{ce}_1 \doteq \mathit{ce}_2 \Rightarrow \mathsf{bool}} \quad \text{SYN_EQ}$$

$\boxed{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{ce} \Leftarrow \tau}$

$$\frac{\begin{array}{l} \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{ce}_1 \Rightarrow \tau \\ \epsilon = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma, x{:}_{\epsilon}\tau \vdash_{\mathit{eff}_0} \mathit{ce}_2 \Leftarrow \tau' \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \textbf{let}\ x = \mathit{ce}_1; \mathit{ce}_2 \Leftarrow \tau'} \quad \text{CHK_LET}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{ce}_1 \Leftarrow \tau_1 \quad .. \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{ce}_n \Leftarrow \tau_n}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} (\mathit{ce}_1, .., \mathit{ce}_n) \Leftarrow (\tau_1, .., \tau_n)} \quad \text{CHK_TUPLE}$$

$$\frac{\begin{array}{l} \epsilon = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\epsilon} \mathit{ce}_1 \Rightarrow (\tau_1, .., \tau_n) \\ \Gamma' = x_1 : \tau_1[\epsilon], .., x_n : \tau_n[\epsilon] \\ \Sigma_c; \Gamma, \Gamma' \vdash_{\mathit{eff}_0} \mathit{ce}_2 \Leftarrow \tau' \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \textbf{let}\,(x_1, .., x_n) = \mathit{ce}_1; \mathit{ce}_2 \Leftarrow \tau'} \quad \text{CHK_LETTUPLE}$$

$$\frac{\begin{array}{l} L : \tau\ \textbf{in}\ D(\tau_1, .., \tau_i)\ \textbf{in}\ \Sigma_c \\ \mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} \mathit{ce} \Leftarrow \tau \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} L\ \mathit{ce} \Leftarrow D(\tau_1, .., \tau_i)} \quad \text{CHK_INJECT}$$

$$\frac{\begin{array}{l} \mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} ce \Rightarrow D(\tau_1, .., \tau_i) \\ D(a_1, .., a_k) = \{L_1 : \tau_1' | .. | L_n : \tau_n'\} \in \Sigma_c \\ L_1 : \tau_1'' \,\mathbf{in}\, D(\tau_1, .., \tau_i) \,\mathbf{in}\, \Sigma_c \quad .. \quad L_n : \tau_n'' \,\mathbf{in}\, D(\tau_1, .., \tau_i) \,\mathbf{in}\, \Sigma_c \\ \Sigma_c; \Gamma, x_1{:}_{\mathit{eff}'}\tau_1'' \vdash_{\mathit{eff}_0} ce_1 \Leftarrow \tau \quad .. \quad \Sigma_c; \Gamma, x_n{:}_{\mathit{eff}'}\tau_n'' \vdash_{\mathit{eff}_0} ce_n \Leftarrow \tau \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathbf{case}\, ce\, \mathbf{of}\, \{L_1\, x_1 \to ce_1 | .. | L_n\, x_n \to ce_n\} \Leftarrow \tau} \quad \textsc{Chk_case}$$

$$\frac{\begin{array}{l} \mathsf{type}\ D(a, b) = \{\mathsf{Next}{:}a \mid \mathsf{Done}{:}b\} \in \Sigma_c \\ \Sigma_c; \Gamma \vdash_{\mathbf{0}} ce_1 \Rightarrow A \\ \Sigma_c; \Gamma, x{:}_{\mathbf{0}}A \vdash_{\mathbf{1}} ce_2 \Leftarrow D(A, B) \end{array}}{\Sigma_c; \Gamma \vdash_{\mathbf{1}} \mathbf{iter}\, (x = ce_1)\{ce_2\} \Leftarrow B} \quad \textsc{Chk_iter}$$

$$\frac{\begin{array}{l} \mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} ce_1 \Leftarrow \mathsf{bool} \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce_2 \Leftarrow \tau \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce_3 \Leftarrow \tau \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathbf{if}\, ce_1\, \mathbf{then}\, ce_2\, \mathbf{else}\, ce_3 \Leftarrow \tau} \quad \textsc{Chk_if}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \tau}{\Sigma_c; \Gamma \vdash_{\mathbf{S}} \mathbf{return}\, ce \Leftarrow \mathsf{pred}\ \tau} \quad \textsc{Chk_return}$$

$$\frac{\begin{array}{l} \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce_1 \Rightarrow \mathsf{pred}\ \tau \\ \Sigma_c; \Gamma, x{:}_{\mathbf{S}}\tau \vdash_{\mathbf{S}} ce_2 \Leftarrow \mathsf{pred}\ \tau' \end{array}}{\Sigma_c; \Gamma \vdash_{\mathbf{S}} \mathbf{take}\, x = ce_1; ce_2 \Leftarrow \mathsf{pred}\ \tau'} \quad \textsc{Chk_take}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce \Rightarrow \tau}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} ce \Leftarrow \tau} \quad \textsc{Chk_sub}$$

$\boxed{\Sigma_c \vdash \mathit{cprog}}$

$$\frac{\Sigma_c; \cdot \vdash_{\mathit{eff}} ce \Leftarrow \tau}{\Sigma_c \vdash \mathsf{main} : \tau_{\mathit{eff}} = ce} \quad \textsc{CProg_main}$$

$$\frac{\begin{array}{l} \Sigma_c; \cdot, x{:}_{\mathbf{0}}A \vdash_{\mathbf{0}} ce \Leftarrow B \\ \Sigma_c, \mathsf{fun}\ f(x : A) \to_{\mathbf{0}} B = ce \vdash \mathit{cprog} \end{array}}{\Sigma_c \vdash \mathsf{fun}\ f(x : A) \to_{\mathbf{0}} B = ce \quad \mathit{cprog}} \quad \textsc{CProg_purefun}$$

$$\frac{\begin{array}{l} \Sigma_c, f : A \to_{\mathbf{1}} B; \cdot, x{:}_{\mathbf{0}}A \vdash_{\mathbf{1}} ce \Leftarrow B \\ \Sigma_c, f : A \to_{\mathbf{1}} B \vdash \mathit{cprog} \end{array}}{\Sigma_c \vdash \mathsf{fun}\ f(x : A) \to_{\mathbf{1}} B = ce \quad \mathit{cprog}} \quad \textsc{CProg_impurefun}$$

$$\frac{\begin{array}{l} \Sigma_c, f : \tau \to_{\mathbf{S}} \tau'; \cdot, x{:}_{\mathbf{S}}\tau \vdash_{\mathbf{S}} ce \Leftarrow \tau' \\ \Sigma_c, \mathsf{fun}\ f(x : \tau) \to_{\mathbf{S}} \tau' = ce \vdash \mathit{cprog} \end{array}}{\Sigma_c \vdash \mathsf{fun}\ f(x : \tau) \to_{\mathbf{S}} \tau' = ce \quad \mathit{cprog}} \quad \textsc{CProg_specfun}$$

$$\frac{\begin{array}{l} \Sigma_c \vdash \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}\ \mathsf{ok} \\ \Sigma_c, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\} \vdash \mathit{cprog} \end{array}}{\Sigma_c \vdash \mathbf{sort}\, D(a_1, .., a_i) = \{L_1 : \tau_1 | .. | L_n : \tau_n\} \mathit{cprog}} \quad \textsc{CProg_sortdecl}$$

$$\frac{\begin{array}{l} \Sigma_c \vdash \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}\ \mathsf{ok} \\ \Sigma_c, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\} \vdash \mathit{cprog} \end{array}}{\Sigma_c \vdash \mathbf{type}\, D(a_1, .., a_i) = \{L_1 : A_1 | .. | L_n : A_n\} \mathit{cprog}} \quad \textsc{CProg_typedecl}$$

$\boxed{L : \tau \,\mathbf{in}\, D(\tau_1, .., \tau_i) \,\mathbf{in}\, \Sigma_c}$

$$\frac{\mathbf{sort}\, D(a_1, .., a_i) \,\mathbf{in}\, \Sigma_c \quad L{:}\tau' \in \mathrm{ctors}(D) \in \Sigma_c \quad \tau = \mathbf{subst}\, \tau_1/a_1, .., \tau_i/a_i \,\mathbf{in}\, \tau'}{L : \tau \,\mathbf{in}\, D(\tau_1, .., \tau_i) \,\mathbf{in}\, \Sigma_c} \quad \textsc{Ctor_sort}$$

$$\frac{\textbf{type}\, D(a_1, .., a_i)\, \textbf{in}\, \Sigma_c \quad L{:}A' \in \mathsf{tyctors}(D) \in \Sigma_c \quad A = \textbf{subst}\, A_1/a_1, .., A_i/a_i\, \textbf{in}\, A'}{L : A\, \textbf{in}\, D(A_1, .., A_i)\, \textbf{in}\, \Sigma_c} \quad \textsc{Ctor_type}$$

$\boxed{D(a_1, .., a_k) = \{L_1 : \tau_1 | .. | L_n : \tau_n\} \in \Sigma_c}$

$$\frac{\textbf{sort}\, D(a_1, .., a_k) = \{L_1 : \tau_1 | .. | L_n : \tau_n\}\, \textbf{in}\, \Sigma_c}{D(a_1, .., a_k) = \{L_1 : \tau_1 | .. | L_n : \tau_n\} \in \Sigma_c} \quad \textsc{DL_sort}$$

$$\frac{\textbf{type}\, D(a_1, .., a_k) = \{L_1 : A_1 | .. | L_n : A_n\}\, \textbf{in}\, \Sigma_c}{D(a_1, .., a_k) = \{L_1 : A_1 | .. | L_n : A_n\} \in \Sigma_c} \quad \textsc{DL_type}$$

$\boxed{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se \Rightarrow \tau \rightsquigarrow ce}$

$$\frac{x{:}_{\mathit{eff}'}\tau \in \Gamma \quad \mathit{eff}' \leq \mathit{eff}_0}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} x \Rightarrow \tau \rightsquigarrow x} \quad \textsc{SurfSyn_var}$$

$$\frac{}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{lit} \Rightarrow \mathsf{int} \rightsquigarrow \mathit{lit}} \quad \textsc{SurfSyn_int}$$

$$\frac{}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{blit} \Rightarrow \mathsf{bool} \rightsquigarrow \mathit{blit}} \quad \textsc{SurfSyn_bool}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se \Leftarrow \tau \rightsquigarrow ce}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se : \tau \Rightarrow \tau \rightsquigarrow ce : \tau} \quad \textsc{SurfSyn_annot}$$

$$\frac{\begin{array}{l} \mathit{prim}{:}\tau \rightarrow_{\mathit{eff}'} \tau' \\ \mathit{eff}' \leq \mathit{eff}_0 \\ \mathit{eff}'' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}''} se \Leftarrow \tau \rightsquigarrow ce \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathit{prim}\, se \Rightarrow \tau' \rightsquigarrow \mathit{prim}\, ce} \quad \textsc{SurfSyn_app}$$

$$\frac{\begin{array}{l} f{:}\tau \rightarrow_{\mathit{eff}'} \tau' \in \Sigma_c \\ \mathit{eff}' \leq \mathit{eff}_0 \\ \mathit{eff}'' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}''} se \Leftarrow \tau \rightsquigarrow ce \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} f\, se \Rightarrow \tau' \rightsquigarrow f\, ce} \quad \textsc{SurfSyn_call}$$

$$\frac{\begin{array}{l} A\ \mathsf{eqtype} \\ \mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} se_1 \Rightarrow A \rightsquigarrow ce_1 \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} se_2 \Leftarrow A \rightsquigarrow ce_2 \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se_1 \doteq se_2 \Rightarrow \mathsf{bool} \rightsquigarrow ce_1 \doteq ce_2} \quad \textsc{SurfSyn_eq}$$

$\boxed{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se \Leftarrow \tau \rightsquigarrow ce}$

$$\frac{\begin{array}{l} \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se_1 \Rightarrow \tau \rightsquigarrow ce_1 \\ \epsilon = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} y, \cdot; \mathit{pat} : \tau \rightarrow [\cdot][[\cdot]] se_2 \Leftarrow_\epsilon \tau' \rightsquigarrow ce_2 \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \textbf{let}\, \mathit{pat} = se_1; se_2 \Leftarrow \tau' \rightsquigarrow \textbf{let}\, y = ce_1; ce_2} \quad \textsc{SurfChk_let}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se_1 \Leftarrow \tau_1 \rightsquigarrow ce_1 \quad .. \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se_n \Leftarrow \tau_n \rightsquigarrow ce_n}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} (se_1, .., se_n) \Leftarrow (\tau_1, .., \tau_n) \rightsquigarrow (ce_1, .., ce_n)} \quad \textsc{SurfChk_tuple}$$

$$\frac{\begin{array}{l} L : \tau\, \textbf{in}\, D(\tau_1, .., \tau_i)\, \textbf{in}\, \Sigma_c \\ \mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} se \Leftarrow \tau \rightsquigarrow ce \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} L\, se \Leftarrow D(\tau_1, .., \tau_i) \rightsquigarrow L\, ce} \quad \textsc{SurfChk_inject}$$

$$\frac{\begin{array}{l}\epsilon = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_\epsilon se \Rightarrow \tau \rightsquigarrow ce' \\ M = \overline{pat{:}\tau, \cdot \rightarrow [\cdot][\cdot]\, se} \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} y, \cdot; M \Leftarrow_\epsilon \tau' \rightsquigarrow ce\end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathbf{case}\, se\, \mathbf{of}\, \{pat_1 \rightarrow se_1 | \,..\, | pat_n \rightarrow se_n\} \Leftarrow \tau' \rightsquigarrow \mathbf{let}\, y = ce'; ce} \quad \textsc{SurfChk_case}$$

$$\frac{\begin{array}{l}\Sigma_c; \Gamma \vdash_{\mathbf{0}} se_1 \Rightarrow A \rightsquigarrow ce_1 \\ \mathsf{type}\ D(a, b) = \{\mathsf{Next}{:}a \mid \mathsf{Done}{:}b\} \in \Sigma_c \\ \Sigma_c; \Gamma \vdash_{\mathbf{1}} z, \cdot; pat : A, \cdot \rightarrow [\cdot][[\cdot]] se_2 \Leftarrow_{\mathbf{0}} D(A, B) \rightsquigarrow ce_2\end{array}}{\Sigma_c; \Gamma \vdash_{\mathbf{1}} \mathbf{iter}\, (pat = se_1)\{se_2\} \Leftarrow B \rightsquigarrow \mathbf{iter}\, (z = ce_1)\{ce_2\}} \quad \textsc{SurfChk_iter}$$

$$\frac{\begin{array}{l}\mathit{eff}' = \lfloor \mathit{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}'} se_1 \Leftarrow \mathsf{bool} \rightsquigarrow ce_1 \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se_2 \Leftarrow \tau \rightsquigarrow ce_2 \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se_3 \Leftarrow \tau \rightsquigarrow ce_3\end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \mathbf{if}\, se_1\, \mathbf{then}\, se_2\, \mathbf{else}\, se_3 \Leftarrow \tau \rightsquigarrow \mathbf{if}\, ce_1\, \mathbf{then}\, ce_2\, \mathbf{else}\, ce_3} \quad \textsc{SurfChk_if}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathbf{S}} se \Leftarrow \tau \rightsquigarrow ce}{\Sigma_c; \Gamma \vdash_{\mathbf{S}} \mathbf{return}\, se \Leftarrow \mathsf{pred}\ \tau \rightsquigarrow \mathbf{return}\, ce} \quad \textsc{SurfChk_return}$$

$$\frac{\begin{array}{l}\Sigma_c; \Gamma \vdash_{\mathbf{S}} se_1 \Rightarrow \mathsf{pred}\ \tau \rightsquigarrow ce_1 \\ \Sigma_c; \Gamma \vdash_{\mathbf{S}} y, \cdot; pat : \tau \rightarrow [\cdot][[\cdot]] se_2 \Leftarrow_{\mathbf{S}} \mathsf{pred}\ \tau' \rightsquigarrow ce_2\end{array}}{\Sigma_c; \Gamma \vdash_{\mathbf{S}} \mathbf{take}\, pat = se_1; se_2 \Leftarrow \mathsf{pred}\ \tau' \rightsquigarrow \mathbf{take}\, y = ce_1; ce_2} \quad \textsc{SurfChk_take}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se \Rightarrow \tau \rightsquigarrow ce}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} se \Leftarrow \tau \rightsquigarrow ce} \quad \textsc{SurfChk_sub}$$

$$\boxed{\begin{array}{l}\Sigma_s \vdash prog \\ \quad \rightsquigarrow \Sigma_c \vdash cprog\end{array}}$$

$$\frac{\Sigma_c; \cdot \vdash_{\mathit{eff}} se \Leftarrow \tau \rightsquigarrow ce}{\begin{array}{l}\Sigma_s \vdash \mathsf{main} : \tau_{\mathit{eff}} = se \\ \quad \rightsquigarrow \Sigma_c \vdash \mathsf{main} : \tau_{\mathit{eff}} = ce\end{array}} \quad \textsc{Prog_main}$$

$$\frac{\begin{array}{l}M = \overline{pat{:}A, \cdot \rightarrow [\cdot][\cdot]\, se} \\ \Sigma_c; \cdot \vdash_{\mathbf{0}} y, \cdot; M \Leftarrow_{\mathbf{0}} B \rightsquigarrow ce \\ \Sigma_s, \mathsf{fun}\ f : A \rightarrow_{\mathbf{0}} B = \{\overline{\mathbf{pat} \rightarrow \mathbf{se}}\} \vdash prog \\ \quad \rightsquigarrow \Sigma_c, \mathsf{fun}\ f(y : A) \rightarrow_{\mathbf{0}} B = ce \vdash cprog\end{array}}{\begin{array}{l}\Sigma_s \vdash \mathsf{fun}\ f : A \rightarrow_{\mathbf{0}} B = \{\overline{\mathbf{pat} \rightarrow \mathbf{se}}\} \quad prog \\ \quad \rightsquigarrow \Sigma_c \vdash \mathsf{fun}\ f(y : A) \rightarrow_{\mathbf{0}} B = ce \quad cprog\end{array}} \quad \textsc{Prog_purefun}$$

$$\frac{\begin{array}{l}M = \overline{pat{:}A, \cdot \rightarrow [\cdot][\cdot]\, se} \\ \Sigma_c, f : A \rightarrow_{\mathbf{1}} B; \cdot \vdash_{\mathbf{1}} y, \cdot; M \Leftarrow_{\mathbf{0}} B \rightsquigarrow ce \\ \Sigma_s, f : A \rightarrow_{\mathbf{1}} B \vdash prog \\ \quad \rightsquigarrow \Sigma_c, f : A \rightarrow_{\mathbf{1}} B \vdash cprog\end{array}}{\begin{array}{l}\Sigma_s \vdash \mathsf{fun}\ f : A \rightarrow_{\mathbf{1}} B = \{\overline{\mathbf{pat} \rightarrow \mathbf{se}}\} \quad prog \\ \quad \rightsquigarrow \Sigma_c \vdash \mathsf{fun}\ f(y : A) \rightarrow_{\mathbf{1}} B = ce \quad cprog\end{array}} \quad \textsc{Prog_impurefun}$$

$$\frac{\begin{array}{l}M = \overline{pat{:}\tau, \cdot \rightarrow [\cdot][\cdot]\, se} \\ \Sigma_c, f : \tau \rightarrow_{\mathbf{S}} \tau'; \cdot \vdash_{\mathbf{S}} y, \cdot; M \Leftarrow_{\mathbf{S}} \tau' \rightsquigarrow ce \\ \Sigma_s, \mathsf{fun}\ f : \tau \rightarrow_{\mathbf{S}} \tau' = \{\overline{\mathbf{pat} \rightarrow \mathbf{se}}\} \vdash prog \\ \quad \rightsquigarrow \Sigma_c, \mathsf{fun}\ f(y : \tau) \rightarrow_{\mathbf{S}} \tau' = ce \vdash cprog\end{array}}{\begin{array}{l}\Sigma_s \vdash \mathsf{fun}\ f : \tau \rightarrow_{\mathbf{S}} \tau' = \{\overline{\mathbf{pat} \rightarrow \mathbf{se}}\} \quad prog \\ \quad \rightsquigarrow \Sigma_c \vdash \mathsf{fun}\ f(y : \tau) \rightarrow_{\mathbf{S}} \tau' = ce \quad cprog\end{array}} \quad \textsc{Prog_specfun}$$

$$\frac{\begin{array}{c}\Sigma_c \vdash \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}\ \mathsf{ok} \\ \Sigma_s, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\} \vdash \mathit{prog} \\ \rightsquigarrow \Sigma_c, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\} \vdash \mathit{cprog}\end{array}}{\begin{array}{c}\Sigma_s \vdash \mathbf{sort}\, D(a_1, .., a_i) = \{L_1 : \tau_1 | .. | L_n : \tau_n\}\mathit{prog} \\ \rightsquigarrow \Sigma_c \vdash \mathbf{sort}\, D(a_1, .., a_i) = \{L_1 : \tau_1 | .. | L_n : \tau_n\}\mathit{cprog}\end{array}} \quad \textsc{Prog_sortdecl}$$

$$\frac{\begin{array}{c}\Sigma_c \vdash \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}\ \mathsf{ok} \\ \Sigma_s, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\} \vdash \mathit{prog} \\ \rightsquigarrow \Sigma_c, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\} \vdash \mathit{cprog}\end{array}}{\begin{array}{c}\Sigma_s \vdash \mathbf{type}\, D(a_1, .., a_i) = \{L_1 : A_1 | .. | L_n : A_n\}\mathit{prog} \\ \rightsquigarrow \Sigma_c \vdash \mathbf{type}\, D(a_1, .., a_i) = \{L_1 : A_1 | .. | L_n : A_n\}\mathit{cprog}\end{array}} \quad \textsc{Prog_typedecl}$$

$\boxed{M\ \mathsf{has_con}}$

$$\frac{}{L\, pat : \tau, \beta \to [\Gamma][\mathcal{E}]se \mid M\ \mathsf{has_con}} \quad \textsc{HC_here}$$

$$\frac{M\ \mathsf{has_con}}{\beta \to [\Gamma][\mathcal{E}]se \mid M\ \mathsf{has_con}} \quad \textsc{HC_there}$$

$\boxed{M\ \mathsf{has_tup}}$

$$\frac{}{(pat_1, .., pat_n) : \tau, \beta \to [\Gamma][\mathcal{E}]se \mid M\ \mathsf{has_tup}} \quad \textsc{HT_here}$$

$$\frac{M\ \mathsf{has_tup}}{\beta \to [\Gamma][\mathcal{E}]se \mid M\ \mathsf{has_tup}} \quad \textsc{HT_there}$$

$\boxed{M/var \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'}$

$$\frac{}{\cdot/var \overset{y}{\rightsquigarrow}_{\mathit{eff}} \cdot} \quad \textsc{SV_empty}$$

$$\frac{M/var \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'}{x : \tau, \beta \to [\Gamma][\mathcal{E}]se \mid M/var \overset{y}{\rightsquigarrow}_{\mathit{eff}} \beta \to [\Gamma, x{:}_{\mathit{eff}}\tau][\mathbf{let}\, x = y; \mathcal{E}]se \mid M'} \quad \textsc{SV_cons}$$

$\boxed{M/L{:}\tau \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'}$

$$\frac{}{\cdot/L{:}\tau \overset{y}{\rightsquigarrow}_{\mathit{eff}} \cdot} \quad \textsc{SC_empty}$$

$$\frac{M/L{:}\tau \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'}{L\, pat : \tau', \beta \to [\Gamma][\mathcal{E}]se \mid M/L{:}\tau \overset{y}{\rightsquigarrow}_{\mathit{eff}} pat : \tau, \beta \to [\Gamma][\mathcal{E}]se \mid M'} \quad \textsc{SC_match}$$

$$\frac{L \neq L' \quad M/L{:}\tau \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'}{L'\, pat : \tau', \beta \to [\Gamma][\mathcal{E}]se \mid M/L{:}\tau \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'} \quad \textsc{SC_skip}$$

$$\frac{M/L{:}\tau \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'}{x : \tau', \beta \to [\Gamma][\mathcal{E}]se \mid M/L{:}\tau \overset{y}{\rightsquigarrow}_{\mathit{eff}} z : \tau, \beta \to [\Gamma, x{:}_{\mathit{eff}}\tau'][\mathbf{let}\, x = y; \mathcal{E}]se \mid M'} \quad \textsc{SC_var}$$

$\boxed{M/\mathbf{tup} \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'}$

$$\frac{}{\cdot/\mathbf{tup} \overset{y}{\rightsquigarrow}_{\mathit{eff}} \cdot} \quad \textsc{ET_empty}$$

$$\frac{M/\mathbf{tup} \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'}{(pat_1, .., pat_k) : (\tau_1, .., \tau_k), \beta \to [\Gamma][\mathcal{E}]se \mid M/\mathbf{tup} \overset{y}{\rightsquigarrow}_{\mathit{eff}} pat_1 : \tau_1, .., pat_k : \tau_k, \beta \to [\Gamma][\mathcal{E}]se \mid M'} \quad \text{ET_TUPLE}$$

$$\frac{M/\mathbf{tup} \overset{y}{\rightsquigarrow}_{\mathit{eff}} M'}{x : (\tau_1, .., \tau_k), \beta \to [\Gamma][\mathcal{E}]se \mid M/\mathbf{tup} \overset{y}{\rightsquigarrow}_{\mathit{eff}} z_1 : \tau_1, .., z_k : \tau_k, \beta \to [\Gamma, x{:}_{\mathit{eff}}(\tau_1, .., \tau_k)][\mathbf{let}\, x = y; \mathcal{E}]se \mid M'} \quad \text{ET_VA}$$

$\boxed{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \overline{y}; M \Leftarrow_\epsilon \tau \rightsquigarrow ce}$

$$\frac{\Sigma_c; \Gamma, \Gamma' \vdash_{\mathit{eff}_0} se \Leftarrow \tau \rightsquigarrow ce' \quad ce = \mathcal{E}[ce']}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \cdot; \cdot \to [\Gamma'][\mathcal{E}]se \mid M \Leftarrow_\epsilon \tau \rightsquigarrow ce} \quad \text{Cov_DONE}$$

$$\frac{\begin{array}{l} M/var \overset{y}{\rightsquigarrow}_\epsilon M' \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} \overline{y}; M' \Leftarrow_\epsilon \tau \rightsquigarrow ce \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} y, \overline{y}; M \Leftarrow_\epsilon \tau \rightsquigarrow ce} \quad \text{Cov_VAR}$$

$$\frac{\begin{array}{l} M \ \mathsf{has_con} \\ D(a_1, .., a_k) = \{L_1 : \tau_1' | .. | L_n : \tau_n'\} \in \Sigma_c \\ L_1 : \tau_1'' \,\mathbf{in}\, D(\tau_1, .., \tau_i) \,\mathbf{in}\, \Sigma_c \quad .. \quad L_n : \tau_n'' \,\mathbf{in}\, D(\tau_1, .., \tau_i) \,\mathbf{in}\, \Sigma_c \\ M/L_1{:}\tau_1'' \overset{y}{\rightsquigarrow}_\epsilon M_1 \quad .. \quad M/L_n{:}\tau_n'' \overset{y}{\rightsquigarrow}_\epsilon M_n \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} x_1, \overline{y}; M_1 \Leftarrow_\epsilon \tau \rightsquigarrow ce_1 \quad .. \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} x_n, \overline{y}; M_n \Leftarrow_\epsilon \tau \rightsquigarrow ce_n \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} y, \overline{y}; M \Leftarrow_\epsilon \tau \rightsquigarrow \mathbf{case}\, y \,\mathbf{of}\, \{L_1\, x_1 \to ce_1 | .. | L_n\, x_n \to ce_n\}} \quad \text{Cov_CON}$$

$$\frac{\begin{array}{l} M \ \mathsf{has_tup} \\ M/\mathbf{tup} \overset{y}{\rightsquigarrow}_\epsilon M' \\ \Sigma_c; \Gamma \vdash_{\mathit{eff}_0} y_1, .., y_n, \overline{y}; M' \Leftarrow_\epsilon \tau \rightsquigarrow ce \end{array}}{\Sigma_c; \Gamma \vdash_{\mathit{eff}_0} y, \overline{y}; M \Leftarrow_\epsilon \tau \rightsquigarrow \mathbf{let}\, (y_1, .., y_n) = y; ce} \quad \text{Cov_TUP}$$

$\boxed{\mathcal{E}[ce] \rightsquigarrow ce'}$

$$\frac{}{[\cdot][ce] \rightsquigarrow ce} \quad \text{Fill_HOLE}$$

$$\frac{\mathcal{E}[ce] \rightsquigarrow ce'}{\mathbf{let}\, x = ce''; \mathcal{E}[ce] \rightsquigarrow \mathbf{let}\, x = ce''; ce'} \quad \text{Fill_LET}$$

$\boxed{\Sigma_r \vdash \Delta \ \mathsf{wf}}$

$$\frac{}{\Sigma_r \vdash \cdot \ \mathsf{wf}} \quad \text{RWf_EMPTY}$$

$$\frac{\Sigma_r \vdash \Delta \ \mathsf{wf} \quad \mathit{eff} \leq \mathbf{S}}{\Sigma_r \vdash \Delta, x{:}_{\mathit{eff}}\tau \ \mathsf{wf}} \quad \text{RWf_EFF}$$

$$\frac{\Sigma_r \vdash \Delta \ \mathsf{wf} \quad \Gamma = |\Delta| \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \mathsf{bool}}{\Sigma_r \vdash \Delta, x{:}ce\ [\mathsf{log}]\ \mathsf{wf}} \quad \text{RWf_LOG}$$

$$\frac{\Sigma_r \vdash \Delta \ \mathsf{wf} \quad \Gamma = |\Delta| \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Rightarrow \mathsf{pred}\ \tau \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce' \Leftarrow \tau}{\Sigma_r \vdash \Delta, x{:}ce@ce'\ [\mathsf{res}(u)]\ \mathsf{wf}} \quad \text{RWf_RES}$$

$\boxed{\Gamma = |\Delta|}$

$$\frac{}{\cdot = |\cdot|} \quad \text{Erase_EMPTY}$$

$$\frac{\Gamma = |\Delta|}{\Gamma, x{:}_{\mathit{eff}}\tau = |\Delta, x{:}_{\mathit{eff}}\tau|} \quad \text{Erase_EFF}$$

$$\frac{\Gamma = |\Delta|}{\Gamma = |\Delta, x{:}ce\ [\mathsf{log}]|}\ \textsc{Erase_log}$$

$$\frac{\Gamma = |\Delta|}{\Gamma = |\Delta, x{:}ce@ce'\ [\mathsf{res}(u)]|}\ \textsc{Erase_res}$$

$\boxed{\Delta' = \mathsf{affinize}(\Delta)}$

$$\frac{}{\cdot = \mathsf{affinize}(\cdot)}\ \textsc{Aff_empty}$$

$$\frac{\Delta' = \mathsf{affinize}(\Delta)}{\Delta', x{:}_{\mathit{eff}}\,\tau = \mathsf{affinize}(\Delta, x{:}_{\mathit{eff}}\,\tau)}\ \textsc{Aff_eff}$$

$$\frac{\Delta' = \mathsf{affinize}(\Delta)}{\Delta', x{:}ce\ [\mathsf{log}] = \mathsf{affinize}(\Delta, x{:}ce\ [\mathsf{log}])}\ \textsc{Aff_log}$$

$$\frac{\Delta' = \mathsf{affinize}(\Delta)}{\Delta', x{:}ce@ce'\ [\mathsf{res}(0)] = \mathsf{affinize}(\Delta, x{:}ce@ce'\ [\mathsf{res}(0)])}\ \textsc{Aff_res_used}$$

$$\frac{\Delta' = \mathsf{affinize}(\Delta)}{\Delta', x{:}ce@ce'\ [\mathsf{res}(?)] = \mathsf{affinize}(\Delta, x{:}ce@ce'\ [\mathsf{res}(1)])}\ \textsc{Aff_res_avail}$$

$$\frac{\Delta' = \mathsf{affinize}(\Delta)}{\Delta', x{:}ce@ce'\ [\mathsf{res}(?)] = \mathsf{affinize}(\Delta, x{:}ce@ce'\ [\mathsf{res}(?)])}\ \textsc{Aff_res_opt}$$

$\boxed{\mathsf{zero}(\Delta)}$

$$\frac{}{\mathsf{zero}(\cdot)}\ \textsc{Zero_empty}$$

$$\frac{\mathsf{zero}(\Delta)}{\mathsf{zero}(\Delta, x{:}_{\mathit{eff}}\,\tau)}\ \textsc{Zero_eff}$$

$$\frac{\mathsf{zero}(\Delta)}{\mathsf{zero}(\Delta, x{:}ce\ [\mathsf{log}])}\ \textsc{Zero_log}$$

$$\frac{\mathsf{zero}(\Delta) \quad u \in \{0, ?\}}{\mathsf{zero}(\Delta, x{:}ce@ce'\ [\mathsf{res}(u)])}\ \textsc{Zero_res}$$

$\boxed{\mathcal{C} = \Delta \Rightarrow \mathcal{C}'}$

$$\frac{}{\mathcal{C} = \cdot \Rightarrow \mathcal{C}}\ \textsc{Close_empty}$$

$$\frac{\mathcal{C} = \Delta \Rightarrow \forall x{:}\tau.\,\mathcal{C}'}{\mathcal{C} = \Delta, x{:}_{\mathit{eff}}\,\tau \Rightarrow \mathcal{C}'}\ \textsc{Close_eff}$$

$$\frac{\mathcal{C} = \Delta \Rightarrow ce \Rightarrow \mathcal{C}'}{\mathcal{C} = \Delta, x{:}ce\ [\mathsf{log}] \Rightarrow \mathcal{C}'}\ \textsc{Close_log}$$

$$\frac{\mathcal{C} = \Delta \Rightarrow \mathcal{C}'}{\mathcal{C} = \Delta, x{:}ce@ce'\ [\mathsf{res}(u)] \Rightarrow \mathcal{C}'}\ \textsc{Close_res}$$

$\boxed{\Sigma_r; \Gamma \vdash \mathcal{C}\ \mathsf{wf}}$

$$\frac{}{\Sigma_r; \Gamma \vdash \top\ \mathsf{wf}}\ \textsc{CtWf_top}$$

$$\frac{\Sigma_r; \Gamma \vdash \mathcal{C}_1\ \mathsf{wf} \quad \Sigma_r; \Gamma \vdash \mathcal{C}_2\ \mathsf{wf}}{\Sigma_r; \Gamma \vdash \mathcal{C}_1 \wedge \mathcal{C}_2\ \mathsf{wf}}\ \textsc{CtWf_and}$$

$$\frac{\Sigma_r; \Gamma, x{:}_{\mathbf{S}}\tau \vdash \mathcal{C}\ \mathsf{wf}}{\Sigma_r; \Gamma \vdash \forall x{:}\tau.\, \mathcal{C}\ \mathsf{wf}} \quad \textsc{CtWf_forall}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \mathsf{bool} \quad \Sigma_r; \Gamma \vdash \mathcal{C}\ \mathsf{wf}}{\Sigma_r; \Gamma \vdash ce \Rightarrow \mathcal{C}\ \mathsf{wf}} \quad \textsc{CtWf_impl}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \mathsf{bool}}{\Sigma_r; \Gamma \vdash ce\ \mathsf{wf}} \quad \textsc{CtWf_atom}$$

$$\frac{}{\Sigma_r; \Gamma \vdash \bot\ \mathsf{wf}} \quad \textsc{CtWf_bot}$$

$\boxed{\Sigma_r; \Gamma \vdash_{\mathit{eff}} Pf\ \mathsf{wf}}$

$$\frac{}{\Sigma_r; \Gamma \vdash_{\mathit{eff}} \cdot\ \mathsf{wf}} \quad \textsc{PfWf_nil}$$

$$\frac{\begin{array}{c}\mathit{eff}' \in \{\mathbf{S}, \lfloor \mathit{eff} \rfloor\} \\ \Sigma_r; \Gamma, x{:}_{\mathit{eff}'}\tau \vdash_{\mathit{eff}} Pf\ \mathsf{wf}\end{array}}{\Sigma_r; \Gamma \vdash_{\mathit{eff}} x{:}_{\mathit{eff}'}\tau, Pf\ \mathsf{wf}} \quad \textsc{PfWf_comp}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \mathsf{bool} \quad \Sigma_r; \Gamma \vdash_{\mathit{eff}} Pf\ \mathsf{wf}}{\Sigma_r; \Gamma \vdash_{\mathit{eff}} x{:}ce\ [\mathsf{log}], Pf\ \mathsf{wf}} \quad \textsc{PfWf_log}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce' \Rightarrow \tau \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \mathsf{pred}\ \tau \quad \Sigma_r; \Gamma \vdash_{\mathit{eff}} Pf\ \mathsf{wf}}{\Sigma_r; \Gamma \vdash_{\mathit{eff}} x{:}ce@ce'\ [\mathsf{res}], Pf\ \mathsf{wf}} \quad \textsc{PfWf_res}$$

$\boxed{\Sigma_r \vdash RF\ \mathsf{wf}}$

$$\frac{\Sigma_r; \cdot \vdash_{\mathit{eff}} Pf_1\ \mathsf{wf} \quad \Gamma = \mathsf{Bind}(\cdot; Pf_1) \quad \Sigma_r; \Gamma \vdash_{\mathit{eff}} Pf_2\ \mathsf{wf}}{\Sigma_r \vdash Pf_1 \multimap Pf_2\ [\mathit{eff}]\ \mathsf{wf}} \quad \textsc{RFWf_rfun}$$

$\boxed{\Sigma_r \vdash f{:}RF}$

$$\frac{f{:}RF \in \Sigma_r}{\Sigma_r \vdash f{:}RF} \quad \textsc{RSLk_rfun}$$

$$\frac{f{:}\tau \to_{\mathit{eff}} \tau' \in \Sigma_r}{\Sigma_r \vdash f{:}x{:}_{\mathit{eff}}\tau, \cdot \multimap y{:}_{\mathit{eff}}\tau', \cdot\ [\mathit{eff}]} \quad \textsc{RSLk_sig}$$

$$\frac{f(\mathbf{x} : \tau) \to_{\mathit{eff}} \tau' = ce \in \Sigma_r}{\Sigma_r \vdash f{:}x{:}_{\mathit{eff}}\tau, \cdot \multimap y{:}_{\mathit{eff}}\tau', \cdot\ [\mathit{eff}]} \quad \textsc{RSLk_def}$$

$\boxed{\overline{A} = \{Pf\}}$

$$\frac{}{[] = \{\cdot\}} \quad \textsc{PfTys_nil}$$

$$\frac{\overline{A} = \{Pf\}}{A{::}\overline{A} = \{x{:}_{\mathbf{0}}A, Pf\}} \quad \textsc{PfTys_pure}$$

$$\frac{\overline{A} = \{Pf\}}{\overline{A} = \{x{:}_{\mathbf{S}}\tau, Pf\}} \quad \textsc{PfTys_spec}$$

$$\frac{\overline{A} = \{Pf\}}{\overline{A} = \{x{:}ce\ [\mathsf{log}], Pf\}} \quad \textsc{PfTys_log}$$

$$\frac{\overline{A} = \{Pf\}}{\overline{A} = \{x{:}ce@ce'\ [\mathsf{res}], Pf\}} \quad \textsc{PfTys_res}$$

$\boxed{\Delta = \mathsf{PfToCtx}(\Delta'; Pf)}$

$$\frac{}{\Delta = \mathsf{PfToCtx}(\Delta; \cdot)} \quad \textsc{PfCtx_nil}$$

$$\frac{\Delta = \mathsf{PfToCtx}(\Delta', x{:}_{\mathit{eff}}\tau; Pf)}{\Delta = \mathsf{PfToCtx}(\Delta'; x{:}_{\mathit{eff}}\tau, Pf)} \quad \textsc{PfCtx_comp}$$

$$\frac{\Delta = \mathsf{PfToCtx}(\Delta', x{:}ce\ [\mathsf{log}]; Pf)}{\Delta = \mathsf{PfToCtx}(\Delta'; x{:}ce\ [\mathsf{log}], Pf)} \quad \textsc{PfCtx_log}$$

$$\frac{\Delta = \mathsf{PfToCtx}(\Delta', x{:}ce@ce'\ [\mathsf{res}(1)]; Pf)}{\Delta = \mathsf{PfToCtx}(\Delta'; x{:}ce@ce'\ [\mathsf{res}], Pf)} \quad \textsc{PfCtx_res}$$

$\boxed{\Sigma_c = \mathsf{Comp}(\Sigma_r)}$

$$\frac{}{\cdot = \mathsf{Comp}(\cdot)} \quad \textsc{RSErase_empty}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r)}{\Sigma_c, f : F = \mathsf{Comp}(\Sigma_r, f : F)} \quad \textsc{RSErase_ext}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r)}{\Sigma_c, \mathsf{fun}\ f(x : \tau) \to_{\mathit{eff}} \tau' = ce = \mathsf{Comp}(\Sigma_r, \mathsf{fun}\ f(x : \tau) \to_{\mathit{eff}} \tau' = ce)} \quad \textsc{RSErase_fundef}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \overline{A}_1 = \{Pf_1\} \quad A = \mathsf{Prod}(\overline{A}_1) \quad \overline{A}_2 = \{Pf_2\} \quad B = \mathsf{Prod}(\overline{A}_2)}{\Sigma_c, f : A \to_{\mathit{eff}} B = \mathsf{Comp}(\Sigma_r, f : Pf_1 \multimap Pf_2\ [\mathit{eff}])} \quad \textsc{RSErase_rfun}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r)}{\Sigma_c, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\} = \mathsf{Comp}(\Sigma_r, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\})} \quad \textsc{RSErase_sort}$$

$$\frac{\Sigma_c = \mathsf{Comp}(\Sigma_r)}{\Sigma_c, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\} = \mathsf{Comp}(\Sigma_r, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\})} \quad \textsc{RSErase_type}$$

$\boxed{\vdash \Sigma_r\ \mathsf{wf}}$

$$\frac{}{\vdash \cdot\ \mathsf{wf}} \quad \textsc{RSWf_empty}$$

$$\frac{\vdash \Sigma_r\ \mathsf{wf}}{\vdash \Sigma_r, f : F\ \mathsf{wf}} \quad \textsc{RSWf_ext}$$

$$\frac{\vdash \Sigma_r\ \mathsf{wf} \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma, x{:}_{\mathit{eff}}\tau \vdash_{\mathit{eff}} ce \Leftarrow \tau'}{\vdash \Sigma_r, \mathsf{fun}\ f(x : \tau) \to_{\mathit{eff}} \tau' = ce\ \mathsf{wf}} \quad \textsc{RSWf_fundef}$$

$$\frac{\vdash \Sigma_r\ \mathsf{wf} \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_r \vdash RF\ \mathsf{wf}}{\vdash \Sigma_r, f : RF\ \mathsf{wf}} \quad \textsc{RSWf_rfun}$$

$\boxed{\Sigma_r; \Delta \vdash \mathit{lpf} \Rightarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{x{:}ce\ [\mathsf{log}] \in \Delta}{\Sigma_r; \Delta \vdash x \Rightarrow ce \dashv \Delta \rightsquigarrow \top} \quad \textsc{LSyn_var}$$

$$\frac{\begin{array}{l}\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad f(\mathbf{x} : \tau) \to_{\mathit{eff}} \tau' = ce' \in \Sigma_c \quad \mathit{eff} \leq \mathbf{S} \\ \Gamma = |\Delta| \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \tau\end{array}}{\Sigma_r; \Delta \vdash \mathsf{unfold}\ f(ce) \Rightarrow (f\ ce) \doteq [ce/x]ce' \dashv \Delta \rightsquigarrow \top} \quad \textsc{LSyn_unfold}$$

$$\frac{\Sigma_r; \Delta \vdash \mathit{lpf} \Leftarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash \mathit{lpf} : ce \Rightarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{LSyn_annot}$$

$$\frac{\Sigma_r; \Delta \vdash rpf \Rightarrow \mathbf{return}\, ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash \mathsf{open\text{-}ret}\ rpf \Rightarrow ce \doteq ce' \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{LSyn_openret}$$

$\boxed{\Sigma_r; \Delta \vdash lpf \Leftarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{}{\Sigma_r; \Delta \vdash \mathsf{auto} \Leftarrow ce \dashv \Delta \rightsquigarrow ce} \quad \textsc{LChk_auto}$$

$$\frac{\Sigma_r; \Delta \vdash lpf \Rightarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash lpf \Leftarrow ce' \dashv \Delta' \rightsquigarrow \mathcal{C} \wedge (ce \Rightarrow (ce'))} \quad \textsc{LChk_sub}$$

$\boxed{\Sigma_r; \Delta \vdash x : ce@ce' \dashv \Delta'}$

$$\frac{x \neq y \quad \Sigma_r; \Delta \vdash x : ce@ce' \dashv \Delta'}{\Sigma_r; \Delta, y{:}_{\mathit{eff}}\tau \vdash x : ce@ce' \dashv \Delta', y{:}_{\mathit{eff}}\tau} \quad \textsc{RLk_there_eff}$$

$$\frac{x \neq y \quad \Sigma_r; \Delta \vdash x : ce@ce' \dashv \Delta'}{\Sigma_r; \Delta, y{:}ce''\ [\mathsf{log}] \vdash x : ce@ce' \dashv \Delta', y{:}ce''\ [\mathsf{log}]} \quad \textsc{RLk_there_log}$$

$$\frac{x \neq y \quad \Sigma_r; \Delta \vdash x : ce@ce' \dashv \Delta'}{\Sigma_r; \Delta, y{:}ce''@ce'''\ [\mathsf{res}(u)] \vdash x : ce@ce' \dashv \Delta', y{:}ce''@ce'''\ [\mathsf{res}(u)]} \quad \textsc{RLk_there_res}$$

$$\frac{u \in \{1, ?\}}{\Sigma_r; \Delta, x{:}ce@ce'\ [\mathsf{res}(u)] \vdash x : ce@ce' \dashv \Delta, x{:}ce@ce'\ [\mathsf{res}(0)]} \quad \textsc{RLk_here}$$

$\boxed{\Sigma_r; \Delta \vdash rpf \Rightarrow ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{\Sigma_r; \Delta \vdash x : ce@ce' \dashv \Delta'}{\Sigma_r; \Delta \vdash x \Rightarrow ce@ce' \dashv \Delta' \rightsquigarrow \top} \quad \textsc{RSyn_var}$$

$$\frac{\Sigma_r; \Delta \vdash rpf \Leftarrow ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash rpf : ce@ce' \Rightarrow ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{RSyn_annot}$$

$\boxed{\Sigma_r; \Delta \vdash rpf \Leftarrow ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{\Sigma_r; \Delta \vdash lpf \Leftarrow ce_1 \doteq ce_2 \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash \mathsf{make\text{-}ret}\ lpf \Leftarrow \mathbf{return}\, ce_1@ce_2 \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{RChk_makeret}$$

$$\frac{\Sigma_r; \Delta \vdash_{\mathit{eff}} crt \Leftarrow x{:}_{\mathbf{S}}\tau, y{:}ce_1@x\ [\mathsf{res}], z{:}ce_2@ce_3\ [\mathsf{res}], \cdot \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash \mathsf{make\text{-}take}\ crt \Leftarrow \mathbf{take}\, x = ce_1; ce_2@ce_3 \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{RChk_maketake}$$

$$\frac{\Sigma_r; \Delta \vdash rpf \Rightarrow ce_1@ce_1' \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash rpf \Leftarrow ce_2@ce_2' \dashv \Delta' \rightsquigarrow (\mathcal{C} \wedge (ce_1 \doteq ce_2)) \wedge (ce_1' \doteq ce_2')} \quad \textsc{RChk_sub}$$

$\boxed{\Sigma_r; \Delta \vdash Pf_1 = Pf_2 \rightsquigarrow \mathcal{C}}$

$$\frac{}{\Sigma_r; \Delta \vdash \cdot = \cdot \rightsquigarrow \top} \quad \textsc{PfEq_nil}$$

$$\frac{\Sigma_r; \Delta, x{:}_{\mathit{eff}}\tau \vdash Pf_1 = [x/y]Pf_2 \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash x{:}_{\mathit{eff}}\tau, Pf_1 = y{:}_{\mathit{eff}}\tau, Pf_2 \rightsquigarrow \forall x{:}\tau.\, \mathcal{C}} \quad \textsc{PfEq_comp}$$

$$\frac{\Sigma_r; \Delta \vdash Pf_1 = Pf_2 \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash x{:}ce_1\ [\mathsf{log}], Pf_1 = y{:}ce_2\ [\mathsf{log}], Pf_2 \rightsquigarrow ((ce_1 \doteq ce_2)) \wedge \mathcal{C}} \quad \textsc{PfEq_log}$$

$$\frac{\Sigma_r; \Delta \vdash Pf_1 = Pf_2 \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash x{:}ce_1@ce_1'\ [\mathsf{res}], Pf_1 = y{:}ce_2@ce_2'\ [\mathsf{res}], Pf_2 \rightsquigarrow (((ce_1 \doteq ce_2)) \wedge (ce_1' \doteq ce_2')) \wedge \mathcal{C}} \quad \textsc{PfEq_res}$$

$\boxed{\Sigma_r \vdash \Delta_0 \sqcap \Delta_1 = \Delta}$

$$\frac{}{\Sigma_r \vdash \cdot \sqcap \cdot = \cdot} \quad \textsc{Merge_empty}$$

$$\frac{\Sigma_r \vdash \Delta_0 \sqcap \Delta_1 = \Delta}{\Sigma_r \vdash \Delta_0, x{:}_{\mathit{eff}}\tau \sqcap \Delta_1, x{:}_{\mathit{eff}}\tau = \Delta, x{:}_{\mathit{eff}}\tau} \quad \textsc{Merge_eff}$$

$$\frac{\Sigma_r \vdash \Delta_0 \sqcap \Delta_1 = \Delta}{\Sigma_r \vdash \Delta_0, x{:}ce\ [\mathsf{log}] \sqcap \Delta_1, x{:}ce\ [\mathsf{log}] = \Delta, x{:}ce\ [\mathsf{log}]} \quad \textsc{Merge_log}$$

$$\frac{\Sigma_r \vdash \Delta_0 \sqcap \Delta_1 = \Delta \quad u = u_0 \sqcap u_1}{\Sigma_r \vdash \Delta_0, x{:}ce@ce'\ [\mathsf{res}(u_0)] \sqcap \Delta_1, x{:}ce@ce'\ [\mathsf{res}(u_1)] = \Delta, x{:}ce@ce'\ [\mathsf{res}(u)]} \quad \textsc{Merge_res}$$

$\boxed{\Sigma_r; \Gamma \vdash q : \mathit{Pf} \dashv \Delta}$

$$\frac{}{\Sigma_r; \Gamma \vdash () : \cdot \dashv \cdot} \quad \textsc{RPM_empty}$$

$$\frac{\Sigma_r; \Gamma, x{:}_{\mathit{eff}}\tau \vdash q : [x/y]\mathit{Pf} \dashv \Delta}{\Sigma_r; \Gamma \vdash (x, q) : y{:}_{\mathit{eff}}\tau, \mathit{Pf} \dashv \cdot, x{:}_{\mathit{eff}}\tau, \Delta} \quad \textsc{RPM_comp}$$

$$\frac{\Sigma_r; \Gamma \vdash q : [x/y]\mathit{Pf} \dashv \Delta}{\Sigma_r; \Gamma \vdash (x, q) : y{:}ce\ [\mathsf{log}], \mathit{Pf} \dashv \cdot, x{:}ce\ [\mathsf{log}], \Delta} \quad \textsc{RPM_log}$$

$$\frac{\Sigma_r; \Gamma \vdash q : [x/y]\mathit{Pf} \dashv \Delta}{\Sigma_r; \Gamma \vdash (x, q) : y{:}ce@ce'\ [\mathsf{res}], \mathit{Pf} \dashv \cdot, x{:}ce@ce'\ [\mathsf{res}(1)], \Delta} \quad \textsc{RPM_res}$$

$\boxed{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{crt} \Rightarrow \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{\begin{array}{l}\Gamma = |\Delta| \quad \Sigma_r; \Gamma \vdash_{\mathit{eff}} \mathit{Pf}\ \mathsf{wf} \\ \Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{crt} \Leftarrow \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{crt} : \mathit{Pf} \Rightarrow \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{RCSyn_annot}$$

$$\frac{\begin{array}{l}\Sigma_r \vdash f{:}RF \quad \mathit{eff}' \in \{\mathbf{S}, \lfloor \mathit{eff} \rfloor\} \\ \Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{rsp} : RF \gg \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} f\ \mathit{rsp} \Rightarrow \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{RCSyn_call}$$

$$\frac{\begin{array}{l}\mathit{eff}' \in \{\mathbf{S}, \lfloor \mathit{eff} \rfloor\} \\ \Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{rsp} : RF \gg \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{prim}\ \mathit{rsp} \Rightarrow \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{RCSyn_primapp}$$

$$\frac{\Sigma_r; \Delta \vdash \mathit{rpf} \Rightarrow \mathbf{take}\, x = ce_1; ce_2@ce_3 \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathsf{open\text{-}take}\ \mathit{rpf} \Rightarrow x{:}_{\mathbf{S}}\tau, y{:}ce_1@x\ [\mathsf{res}], z{:}ce_2@ce_3\ [\mathsf{res}], \cdot \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{RCSyn_opentake}$$

$\boxed{\Sigma_r; \Delta \vdash_{\mathit{eff}} y{:}ce\ \mathsf{is}\ L\,x \rightarrow \mathit{crt} \Leftarrow \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{\begin{array}{l}\Gamma = |\Delta| \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}} ce \Rightarrow D(\tau_1, .., \tau_i) \\ L : \tau\, \mathbf{in}\, D(\tau_1, .., \tau_i)\, \mathbf{in}\, \Sigma_c \\ \Sigma_r; \Delta, x{:}_{\mathit{eff}}\tau, y{:}ce \doteq L\,x\ [\mathsf{log}] \vdash_{\mathit{eff}} \mathit{crt} \Leftarrow \mathit{Pf} \dashv \Delta', x{:}_{\mathit{eff}}\tau, y{:}ce \doteq L\,x\ [\mathsf{log}] \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} y{:}ce\ \mathsf{is}\ L\,x \rightarrow \mathit{crt} \Leftarrow \mathit{Pf} \dashv \Delta' \rightsquigarrow \forall x{:}\tau.\ ce \doteq L\,x \Rightarrow \mathcal{C}} \quad \textsc{CBr_branch}$$

$\boxed{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{crt} \Leftarrow \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}}$

$$\frac{\begin{array}{l}\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{crt}_1 \Rightarrow \mathit{Pf}' \dashv \Delta' \rightsquigarrow \mathcal{C}_1 \\ \Gamma = |\Delta'| \quad \Sigma_r; \Gamma \vdash q : \mathit{Pf}' \dashv \Delta'' \\ \Sigma_r; \Delta', \Delta'' \vdash_{\mathit{eff}} \mathit{crt}_2 \Leftarrow \mathit{Pf} \dashv \Delta''', \Delta'''' \rightsquigarrow \mathcal{C}_2 \\ \mathsf{len}(\Delta') = \mathsf{len}(\Delta''') \\ \mathsf{len}(\Delta'') = \mathsf{len}(\Delta'''') \\ \mathsf{zero}(\Delta'''')\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathbf{let}\ q = \mathit{crt}_1; \mathit{crt}_2 \Leftarrow \mathit{Pf} \dashv \Delta''' \rightsquigarrow \mathcal{C}_1 \wedge \mathcal{C}_2} \quad \textsc{RCChk_let}$$

$$\frac{\begin{array}{l} \mathit{eff}' = \lfloor \mathit{eff} \rfloor \\ \Gamma = |\Delta| \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}'} \mathit{ce} \Leftarrow \mathsf{bool} \\ \Sigma_r; \Delta, x{:}\mathit{ce} \doteq \mathsf{true}\ [\mathsf{log}] \vdash_{\mathit{eff}'} \mathit{crt}_1 \Leftarrow \mathit{Pf} \dashv \Delta_1, x{:}\mathit{ce} \doteq \mathsf{true}\ [\mathsf{log}] \rightsquigarrow \mathcal{C}_1 \\ \Sigma_r; \Delta, x{:}\mathit{ce} \doteq \mathsf{false}\ [\mathsf{log}] \vdash_{\mathit{eff}'} \mathit{crt}_2 \Leftarrow \mathit{Pf} \dashv \Delta_2, x{:}\mathit{ce} \doteq \mathsf{false}\ [\mathsf{log}] \rightsquigarrow \mathcal{C}_2 \\ \Sigma_r \vdash \Delta_1 \sqcap \Delta_2 = \Delta' \end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathbf{if}\ [x]\mathit{ce}\ \mathbf{then}\ \mathit{crt}_1\ \mathbf{else}\ \mathit{crt}_2 \Leftarrow \mathit{Pf} \dashv \Delta' \rightsquigarrow (\mathit{ce} \doteq \mathsf{true} \Rightarrow \mathcal{C}_1) \wedge (\mathit{ce} \doteq \mathsf{false} \Rightarrow \mathcal{C}_2)} \quad \textsc{RCChk_if}$$

$$\frac{\begin{array}{l} \mathit{eff}' = \lfloor \mathit{eff} \rfloor \\ \Gamma = |\Delta| \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}'} \mathit{ce} \Rightarrow D(\tau_1, \ldots, \tau_i) \\ D(a_1, \ldots, a_k) = \{L_1 : \tau_1' | \ldots | L_n : \tau_n'\} \in \Sigma_c \\ \Sigma_r; \Delta \vdash_{\mathit{eff}'} y{:}\mathit{ce}\ \mathsf{is}\ L_1\ x_1 \to \mathit{crt}_1 \Leftarrow \mathit{Pf} \dashv \Delta_1 \rightsquigarrow \mathcal{C}_1 \quad \ldots \quad \Sigma_r; \Delta \vdash_{\mathit{eff}'} y{:}\mathit{ce}\ \mathsf{is}\ L_n\ x_n \to \mathit{crt}_n \Leftarrow \mathit{Pf} \dashv \Delta_n \rightsquigarrow \mathcal{C}_n \\ \Sigma_r \vdash \sqcap(\Delta_1; \ldots; \Delta_n) = \Delta' \end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathbf{case}\ [y]\mathit{ce}\ \mathbf{of}\ \{L_1\ x_1 \to \mathit{crt}_1 | \ldots | L_n\ x_n \to \mathit{crt}_n\} \Leftarrow \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}_1 \wedge \mathcal{C}_n} \quad \textsc{RC}$$

$$\frac{\begin{array}{l} \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \mathsf{type}\ D(a, b) = \{\mathsf{Next}{:}a \mid \mathsf{Done}{:}b\} \in \Sigma_c \\ \Gamma = |\Delta| \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} \mathit{ce} \Rightarrow \mathsf{pred}\ D(A, B) \\ \Sigma_r; \Delta \vdash_{\mathbf{0}} \mathit{crt}_1 \Leftarrow x{:}_{\mathbf{0}}A, y{:}\mathit{ce}@\mathit{Next}\ x\ [\mathsf{res}], \cdot \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_r; \Delta', x{:}_{\mathbf{0}}A, y{:}\mathit{ce}@\mathit{Next}\ x\ [\mathsf{res}(1)] \vdash_{\mathbf{1}} \mathit{crt}_2 \Leftarrow z{:}_{\mathbf{0}}D(A, B), y{:}\mathit{ce}@z\ [\mathsf{res}], \cdot \dashv \Delta', x{:}_{\mathbf{0}}A, y{:}\mathit{ce}@\mathit{Next}\ x\ [\mathsf{res}(1)] \rightsquigarrow \mathcal{C}' \end{array}}{\Sigma_r; \Delta \vdash_{\mathbf{1}} \mathbf{iter}\ [\mathit{ce}](q = \mathit{crt}_1)\{\mathit{crt}_2\} \Leftarrow z{:}_{\mathbf{0}}B, y{:}\mathit{ce}@\mathit{Done}\ z\ [\mathsf{res}], \cdot \dashv \Delta' \rightsquigarrow \mathcal{C} \wedge \forall x{:}A.\mathcal{C}'}$$

$$\frac{\Delta' = \mathsf{affinize}(\Delta)}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathsf{exfalso} \Leftarrow \mathit{Pf} \dashv \Delta' \rightsquigarrow \bot} \quad \textsc{RCChk_exfalso}$$

$$\frac{\begin{array}{l} \Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{crt} \Rightarrow \mathit{Pf}' \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_r; \Delta' \vdash \mathit{Pf}' = \mathit{Pf} \rightsquigarrow \mathcal{C}' \end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{crt} \Leftarrow \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \textsc{RCChk_sub}$$

$$\boxed{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{rsp} : \mathit{RF} \gg \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}}$$

$$\frac{}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \cdot : \cdot \multimap \mathit{Pf}\ [\mathit{eff}] \gg \mathit{Pf} \dashv \Delta \rightsquigarrow \top} \quad \textsc{Spine_nil}$$

$$\frac{\begin{array}{l} \Gamma = |\Delta| \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}} \mathit{ce} \Leftarrow \tau \\ \Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{rsp} : [\mathit{ce}/x]\mathit{Pf}_1 \multimap [\mathit{ce}/x]\mathit{Pf}_2\ [\mathit{eff}] \gg \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C} \end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{ce}, \mathit{rsp} : x{:}_{\mathit{eff}}\tau, \mathit{Pf}_1 \multimap \mathit{Pf}_2\ [\mathit{eff}] \gg \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{Spine_core}$$

$$\frac{\begin{array}{l} \Sigma_r; \Delta \vdash \mathit{lpf} \Leftarrow \mathit{ce} \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_r; \Delta' \vdash_{\mathit{eff}} \mathit{rsp} : \mathit{Pf}_1 \multimap \mathit{Pf}_2\ [\mathit{eff}] \gg \mathit{Pf} \dashv \Delta'' \rightsquigarrow \mathcal{C}' \end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{lpf}, \mathit{rsp} : x{:}\mathit{ce}\ [\mathsf{log}], \mathit{Pf}_1 \multimap \mathit{Pf}_2\ [\mathit{eff}] \gg \mathit{Pf} \dashv \Delta'' \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \textsc{Spine_log}$$

$$\frac{\begin{array}{l} \Sigma_r; \Delta \vdash \mathit{rpf} \Leftarrow \mathit{ce}_1@\mathit{ce}_2 \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_r; \Delta' \vdash_{\mathit{eff}} \mathit{rsp} : \mathit{Pf}_1 \multimap \mathit{Pf}_2\ [\mathit{eff}] \gg \mathit{Pf} \dashv \Delta'' \rightsquigarrow \mathcal{C}' \end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{rpf}, \mathit{rsp} : x{:}\mathit{ce}_1@\mathit{ce}_2\ [\mathsf{res}], \mathit{Pf}_1 \multimap \mathit{Pf}_2\ [\mathit{eff}] \gg \mathit{Pf} \dashv \Delta'' \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \textsc{Spine_res}$$

$$\boxed{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{rsp} : \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}}$$

$$\frac{}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \cdot : \cdot \dashv \Delta \rightsquigarrow \top} \quad \textsc{Tup_nil}$$

$$\frac{\begin{array}{l} \Gamma = |\Delta| \quad \Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathit{eff}} \mathit{ce} \Leftarrow \tau \\ \Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{rsp} : [\mathit{ce}/x]\mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C} \end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{ce}, \mathit{rsp} : x{:}_{\mathit{eff}}\tau, \mathit{Pf} \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \textsc{Tup_core}$$

$$\frac{\begin{array}{l} \Sigma_r; \Delta \vdash \mathit{lpf} \Leftarrow \mathit{ce} \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_r; \Delta' \vdash_{\mathit{eff}} \mathit{rsp} : \mathit{Pf} \dashv \Delta'' \rightsquigarrow \mathcal{C}' \end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} \mathit{lpf}, \mathit{rsp} : x{:}\mathit{ce}\ [\mathsf{log}], \mathit{Pf} \dashv \Delta'' \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \textsc{Tup_log}$$

$$\frac{\begin{array}{c}\Sigma_r; \Delta \vdash rpf \Leftarrow ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_r; \Delta' \vdash_{\mathit{eff}} rsp : Pf \dashv \Delta'' \rightsquigarrow \mathcal{C}'\end{array}}{\Sigma_r; \Delta \vdash_{\mathit{eff}} rpf, rsp : x{:}ce@ce' \ [\mathsf{res}], Pf \dashv \Delta'' \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \textsc{Tup_res}$$

$\boxed{\Sigma_r \vdash rprog \dashv \Sigma_r' \rightsquigarrow \mathcal{C}}$

$$\frac{\Sigma_r; \cdot \vdash_{\mathit{eff}} crt \Leftarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r \vdash \mathsf{main} : Pf_{\mathit{eff}} = crt \dashv \Sigma_r \rightsquigarrow \mathcal{C}} \quad \textsc{RProg_main}$$

$$\frac{\begin{array}{c}\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c \vdash \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\}\ \mathsf{ok} \\ \Sigma_r, \mathsf{sort}\ D(\overline{a}) = \{\overline{L : \tau}\} \vdash rprog \dashv \Sigma_r' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r \vdash \mathbf{sort}\, D(a_1, .., a_i) = \{L_1 : \tau_1 | .. | L_n : \tau_n\} rprog \dashv \Sigma_r' \rightsquigarrow \mathcal{C}} \quad \textsc{RProg_sortdecl}$$

$$\frac{\begin{array}{c}\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c \vdash \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\}\ \mathsf{ok} \\ \Sigma_r, \mathsf{type}\ D(\overline{a}) = \{\overline{L : A}\} \vdash rprog \dashv \Sigma_r' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r \vdash \mathbf{type}\, D(a_1, .., a_i) = \{L_1 : A_1 | .. | L_n : A_n\} rprog \dashv \Sigma_r' \rightsquigarrow \mathcal{C}} \quad \textsc{RProg_typedecl}$$

$$\frac{\begin{array}{c}\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c; \cdot, x{:}_{\mathbf{0}} A \vdash_{\mathbf{0}} ce \Leftarrow B \\ \Sigma_r, \mathsf{fun}\ f(x : A) \rightarrow_{\mathbf{0}} B = ce \vdash rprog \dashv \Sigma_r' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r \vdash \mathsf{fun}\ f(x : A) \rightarrow_{\mathbf{0}} B = ce \quad rprog \dashv \Sigma_r' \rightsquigarrow \mathcal{C}} \quad \textsc{RProg_purefun}$$

$$\frac{\begin{array}{c}\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c, f : A \rightarrow_{\mathbf{1}} B; \cdot, x{:}_{\mathbf{0}} A \vdash_{\mathbf{1}} ce \Leftarrow B \\ \Sigma_r, f : A \rightarrow_{\mathbf{1}} B \vdash rprog \dashv \Sigma_r' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r \vdash \mathsf{fun}\ f(x : A) \rightarrow_{\mathbf{1}} B = ce \quad rprog \dashv \Sigma_r' \rightsquigarrow \mathcal{C}} \quad \textsc{RProg_impurefun}$$

$$\frac{\begin{array}{c}\Sigma_c = \mathsf{Comp}(\Sigma_r) \quad \Sigma_c, f : \tau \rightarrow_{\mathbf{S}} \tau'; \cdot, x{:}_{\mathbf{S}} \tau \vdash_{\mathbf{S}} ce \Leftarrow \tau' \\ \Sigma_r, \mathsf{fun}\ f(x : \tau) \rightarrow_{\mathbf{S}} \tau' = ce \vdash rprog \dashv \Sigma_r' \rightsquigarrow \mathcal{C}\end{array}}{\Sigma_r \vdash \mathsf{fun}\ f(x : \tau) \rightarrow_{\mathbf{S}} \tau' = ce \quad rprog \dashv \Sigma_r' \rightsquigarrow \mathcal{C}} \quad \textsc{RProg_specfun}$$

$$\frac{\begin{array}{c}\Sigma_r \vdash Pf_1 \multimap Pf_2\ [\mathbf{0}]\ \mathsf{wf} \\ \Delta = \mathsf{PfToCtx}(\cdot; Pf_1) \\ \Sigma_r; \Delta \vdash_{\mathbf{0}} crt \Leftarrow Pf_2 \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_r, f : Pf_1 \multimap Pf_2\ [\mathbf{0}] \vdash rprog \dashv \Sigma_r' \rightsquigarrow \mathcal{C}'\end{array}}{\Sigma_r \vdash \mathsf{fun}\ f(Pf_1) \rightarrow_{\mathbf{0}} Pf_2 = crt \quad rprog \dashv \Sigma_r' \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \textsc{RProg_purerefined}$$

$$\frac{\begin{array}{c}\Sigma_r \vdash Pf_1 \multimap Pf_2\ [\mathbf{1}]\ \mathsf{wf} \\ \Delta = \mathsf{PfToCtx}(\cdot; Pf_1) \\ \Sigma_r, f : Pf_1 \multimap Pf_2\ [\mathbf{1}]; \Delta \vdash_{\mathbf{1}} crt \Leftarrow Pf_2 \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_r, f : Pf_1 \multimap Pf_2\ [\mathbf{1}] \vdash rprog \dashv \Sigma_r' \rightsquigarrow \mathcal{C}'\end{array}}{\Sigma_r \vdash \mathsf{fun}\ f(Pf_1) \rightarrow_{\mathbf{1}} Pf_2 = crt \quad rprog \dashv \Sigma_r' \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \textsc{RProg_impurerefined}$$

$$\frac{\begin{array}{c}\Sigma_r \vdash Pf_1 \multimap Pf_2\ [\mathbf{S}]\ \mathsf{wf} \\ \Delta = \mathsf{PfToCtx}(\cdot; Pf_1) \\ \Sigma_r, f : Pf_1 \multimap Pf_2\ [\mathbf{S}]; \Delta \vdash_{\mathbf{S}} crt \Leftarrow Pf_2 \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_r, f : Pf_1 \multimap Pf_2\ [\mathbf{S}] \vdash rprog \dashv \Sigma_r' \rightsquigarrow \mathcal{C}'\end{array}}{\Sigma_r \vdash \mathsf{fun}\ f(Pf_1) \rightarrow_{\mathbf{S}} Pf_2 = crt \quad rprog \dashv \Sigma_r' \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \textsc{RProg_specrefined}$$

$\boxed{\Sigma_r \vdash \sqcap \overline{\Delta} = \Delta}$

$$\frac{}{\Sigma_r \vdash \sqcap \Delta_0; \cdot = \Delta_0} \quad \textsc{NMerge_base}$$

$$\frac{\Sigma_r \vdash \Delta_0 \sqcap \Delta_1 = \Delta' \quad \Sigma_r \vdash \sqcap(\Delta'; \overline{\Delta}) = \Delta}{\Sigma_r \vdash \sqcap(\Delta_0; \Delta_1; \overline{\Delta}) = \Delta} \quad \textsc{NMerge_step}$$

```
Definition rules:          234 good    0 bad
Definition rule clauses:   516 good    0 bad
```